

Are galaxies in compact groups special?

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ABSTRACT

Context. Compact groups combine high projected densities with low velocity dispersions, but many are embedded in larger structures. Their significance depends on whether their galaxies differ from comparable regular-group members, not only from field galaxies.

Aims. We test whether galaxies in compact groups of four members (CG₄) remain special against regular groups of four members (RG₄), the four brightest eligible members of regular groups (Control_{4B}), and projected compact cores formed by the brightest group galaxy and its three nearest projected companions (Control_{4C}).

Methods. From previous samples, we retain 62 non-split CG₄ systems. We combine Galaxy Zoo 1 vote fractions, Sloan Digital Sky Survey stellar masses and specific star-formation rates, optical colours, emission-line diagnostics, and seeing-corrected half-light radii. We compare each control separately with covariate-adjusted models, galaxy- and group-level matching, and within-host analyses.

Results. The clearest difference is morphological. CG₄ galaxies, especially satellites, have higher elliptical-vote and lower spiral-vote fractions than regular-group analogues. After accounting for galaxy and group properties, the excess is strongest against Control_{4B} and RG₄ and smallest against Control_{4C}. Matched CG₄ quartets likewise contain more elliptical-vote satellites than their controls, most clearly against Control_{4B} and RG₄ and least clearly against Control_{4C}; adding satellite mass to the match preserves this ordering. No additional within-host excess is detected after accounting for host-centric radius. The morphology result is stable to vote-threshold, vote-fraction, Sérsic-index, and close-pair checks, and is attenuated by a projected pairwise tidal-density proxy. Star-formation, colour, active-galactic-nucleus, magnitude-gap, H α , and fixed-mass size diagnostics show no comparably robust CG₄ offsets.

Conclusions. CG₄ systems therefore host a population shifted towards early-type-like morphologies, consistent with accumulated transformation in dense group cores and/or preprocessing, without establishing a present-day quenching mechanism unique to compact groups.

Key words. galaxies: groups: general – galaxies: evolution – galaxies: interactions – galaxies: star formation – galaxies: elliptical and lenticular, cD – methods: statistical

1. Introduction

Given their high apparent densities and low velocity dispersions (Hickson 1982; Hickson et al. 1992), compact groups appear to be ideal sites for the frequent tidal encounters and mergers that alter galaxy masses, luminosities, sizes, morphologies, and star-formation rates (e.g., Mamon 1992). Mendes de Oliveira & Hickson (1994) provided the classic report of an excess of early-type galaxies in compact groups relative to the field (building on the photometric morphologies of Hickson et al. 1989 and the merger-remnant expectations of Zepf & Whitmore 1993), in line with the broader morphology–density relation (Dressler 1980); multi-wavelength studies of compact-group galaxies likewise report altered infrared, UV, star-formation, and stellar-population properties relative to looser environments (Johnson et al. 2007; Tzanavaris et al. 2010;

Bitsakis et al. 2011; Coenda et al. 2012), and their H I deficiency has been organised into an evolutionary sequence (Verdes-Montenegro et al. 2001).

The interpretation of such contrasts is nevertheless ambiguous. Projected selection can include chance alignments, and many Hickson-like systems need not be isolated bound entities: apparent compact groups may be transient alignments, dense substructures, or cores embedded in looser groups and richer haloes (Mamon 1986; Díaz-Giménez & Mamon 2010; Díaz-Giménez et al. 2020; Zheng & Shen 2021; Taverna et al. 2023). The relevant observational question is therefore not only whether compact-group galaxies differ from field galaxies, but whether they remain different from galaxies occupying regular-group analogues selected with comparable luminosity, redshift, and membership constraints. Recent structural studies find evidence for transformation in compact groups, especially non-isolated systems (Montaguth et al. 2023, 2025, 2026a,b); Montaguth et al. (2026b) compared non-isolated compact-group galaxies with surrounding host populations, but not

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with ordinary-group quartets reproducing compact-group membership, luminosity-ranking, or projected-core selection.

In our previous work (Tricottet et al. 2025), we found that, once the direct consequences of the compact-group selection are accounted for, compact groups of four galaxies have group-scale properties broadly similar to those of regular groups of four galaxies and to the eligible cores of regular groups. Here we instead ask how the inferred galaxy-population contrast changes across ordinary-group counterfactuals: genuine four-member groups (RG₄), the four brightest eligible members (Control_{4B}), and BGG-centred projected cores (Control_{4C}), the closest compact-configuration analogue but not compactness-matched (Sect. 2.1). We adjust for stellar mass, redshift, galaxy rank, and available group-scale properties, and combine individual-galaxy, group-composition, and shared-host contrasts.

Section 2 describes the samples, the classifications, and the statistical design. Section 3 presents the population contrasts, the matched and within-host comparisons, and the robustness and secondary diagnostics. Section 4 discusses the physical interpretation, and Section 5 concludes.

2. Data and classifications

2.1. Samples

We use the compact-group and control samples constructed in our previous study (Tricottet et al. 2025), whose construction we briefly recall. Galaxy properties are taken from the SDSS DR12 catalogue of Tempel et al. (2017), while regular groups are drawn from the SDSS DR7 group catalogue of Lim et al. (2017); the two catalogues were matched through SDSS object identifiers to define the Lim–Tempel working catalogue. Compact groups are selected from the modified Hickson compact-group catalogue of Díaz-Giménez et al. (2018): reliable HMCs with exactly four members, all present in the Lim–Tempel catalogue, satisfying the volume- and luminosity-complete limits $0.005 \leq z_{\text{BGG}} \leq 0.0452$ and $M_{r,\text{BGG}} \leq -21.81$, with all four galaxies lying within 3 magnitudes of the brightest group galaxy (BGG). In contrast to our previous study, we remove CGs classified as *split* following Zheng & Shen (2021), which reduces the compact-group sample from 78 to 62 groups.

The regular-group comparison samples are built from Lim–Tempel groups satisfying the same redshift and BGG-luminosity limits, with at least 4 members and at least 3 satellites within 3 magnitudes of the BGG. The three controls serve different physical purposes. RG₄ contains regular groups with exactly four members (excluding systems identical to a CG₄) and is the most direct equal-membership comparison; it is substantially looser than CG₄ and comparable to the broadest regular-group core comparison: the median (over groups) of the median pairwise projected member separation is 430 kpc for RG₄ (interquartile range 292–494 kpc, proper, angular-diameter distances) against 143 kpc for CG₄ (114–177 kpc), so RG₄ quartets are typically ~3 times more extended than the compact groups they are compared with. Control_{4B} contains the four brightest eligible members of each regular group and tests whether the CG₄ population resembles the luminous core of an ordinary group; its corresponding median pairwise

separation is 447 kpc. Control_{4C} contains the BGG and its three closest projected companions among group members lying within 3 magnitudes of the BGG. We reconstruct Control_{4C} from the eligible Lim–Tempel parent groups by applying this magnitude criterion before ranking companions by projected distance. It is the most compact-core-like ordinary-group analogue: its median pairwise separation is 293 kpc (IQR 187–436 kpc), intermediate between CG₄ and the looser controls. The CG₄–Control_{4C} overlap coefficient in the compactness score $-\log R_{\text{pair,med}}$ is 0.34, and 18% of CG₄ groups lie above the Control_{4C} 95th percentile in compactness, so Control_{4C} provides common support but is not a compactness-matched sample. Every control group containing at least one CG₄ galaxy is excluded from the control samples, so no galaxy appears on both sides of any comparison; the extent of this compact-group/projected-core overlap is quantified in Appendix C. We also remove the systems associated with Lim group 3688, which appears to be spurious: its single outlying-redshift member ($z = 0.048$, against 0.032–0.034 for the other six) is selected into both our Control_{4B} and Control_{4C} samples and inflates the group velocity dispersions to $\approx 2000 \text{ km s}^{-1}$ (1979 and 2042 km s^{-1} , respectively). After these removals, the control samples contain 698 Control_{4B}, 703 Control_{4C}, and 56 RG₄ groups (2792, 2812, and 224 galaxies); the Control_{4B} and Control_{4C} counts differ because the luminosity-ranked and projected-distance-ranked core definitions select different galaxies and therefore retain different Lim groups after the compact-group exclusion.

We use SDSS DR16 (Ahumada et al. 2020) to retrieve stellar masses, star formation rates (SFRs; Brinchmann et al. 2004; Kauffmann et al. 2003b), and Galaxy Zoo morphologies (Lintott et al. 2008, 2011); Appendix A summarises the principal quantities and table joins, and the full SQL query is provided in `src/-data_loader.py`. Throughout, ‘SDSS non-AGN reference’ denotes the AGN-filtered, post-processed reference sample used for calibration and orientation: spectroscopic galaxies with $0.005 \leq z \leq 0.0452$, extinction-corrected Petrosian $r \leq 17.77$, a valid MPA-JHU stellar mass, and finite Galaxy Zoo vote fractions after the pipeline AGN filter. It serves as a field-dominated reference population, not as a control sample. All projected distances and derived quantities adopt the Planck 2015 cosmology (Planck Collaboration et al. 2016), consistently with our previous paper.

2.2. Star-formation and morphology classes

We classify galaxies from SDSS total sSFRs. The inherited DR12 group catalogues store $\log s\text{SFR} = \text{sfr_tot_p50} - \text{lgm_tot_p50}$; they do not obtain this final quantity from specsfr_tot_p50 . MPA-JHU values of -9999 are missing-value sentinels rather than physical measurements, and the same convention is used for MPA-JHU DR8 target quantities by Zeraatgari et al. (2024). We therefore treat rows with unavailable components, including $\text{sfr_tot_p50} = -9999$, as *missing data*: such galaxies are excluded from the star-formation classification and from every sSFR-based figure and fraction, and are reported as counts only (Table 1). All quenched and star-forming fractions quoted in this paper are computed among classified galaxies only. The missing-sSFR incidence is markedly higher in CG₄ (7% of members, and 10% of BGGs) than in the field se-

lection (1%). An exact-spectrum provenance audit shows that the missing-sSFR rows mix `galSpecExtra` sentinel values with spectra absent from the MPA-JHU `galSpecExtra/galSpecInfo` tables; the available cross-match is therefore insufficient to establish the origin of the missing sSFR estimates. Galaxies with a measured sSFR are split into quenched and star-forming classes with a two-component Gaussian mixture model (GMM) in the $\log M_\star$ – $\log$ sSFR plane; BPT/AGN handling and the exact GMM boundary definition are given in Appendix B (Fig. B.1).

Morphologies are drawn from Galaxy Zoo 1 (Lintott et al. 2008, 2011), a single-question classification in which volunteers assigned each galaxy image to one of several morphological categories. The resulting vote fractions, debiased for the redshift-dependent classification bias following Bamford et al. (2009), are provided in the `zooSpec` table through the columns `p_el_debiased` (fraction voting “elliptical”) and `p_cs_debiased` (fraction voting combined-spiral; Lintott et al. 2011). We classify a galaxy as elliptical if `p_el_debiased` exceeds 0.5, as spiral if `p_cs_debiased` exceeds 0.5, and as “uncertain” only when both vote fractions are finite but neither threshold is exceeded. If either vote fraction is missing after merging Galaxy Zoo onto the compact-group and control catalogues, the row is treated as missing Galaxy Zoo data rather than uncertain. These are vote-fraction classes, not a full structural decomposition. In particular, the catalogue does not provide a clean E/S0 separation, so references below to elliptical or early-type galaxies should be read as Galaxy Zoo 1 elliptical-vote proxies for early-type morphology rather than as a bulge-disc structural classification.

2.3. Galaxy sizes

Our primary galaxy-size measure is the circular half-light radius of the pure Sérsic fit in the r band, $R_{\text{chl},r}$, from the bulge+disk decomposition catalogue of Simard et al. (2011). These are PSF-convolved GIM2D fits with re-derived sky levels and deblending designed with crowded fields in mind – the relevant regime for compact-group photometry – so the fitted sizes are corrected for seeing. The catalogue is keyed to SDSS DR7 identifiers; we bridge our DR16 objects to their DR7 counterparts through the SkyServer `SpecDR7` cross-match table. Rows in which the Simard redshift differs from the catalogue redshift by more than 0.005 are treated as wrong matches and their structural values dropped (9 rows). We also checked whether two distinct catalogue galaxies of the same group resolve to a single DR7 identifier, but there were no such occurrences. The Simard sizes are tabulated in kpc under the catalogue’s own cosmology; we convert them back to angular sizes with the catalogue plate scale and then to proper kpc with the same Planck 2015 conversion as every other projected quantity in this paper.

As a secondary size measure we use the SDSS Petrosian radii `petroR50_r` and `petroR90_r`, which also define the concentration index $C = R_{90,r}/R_{50,r}$. Petrosian radii are not corrected for seeing, and the SDSS Petrosian aperture misses a profile-dependent fraction of the flux, roughly 20% for an $n = 4$ profile, which biases $R_{50,r}$ low for concentrated galaxies (Blanton et al. 2001; Graham et al. 2005) – precisely the elliptical/early-type-like population in which CG_4 is enhanced. The Petrosian measure is nevertheless retained

because it is model-independent, essentially fully available, and provides the like-for-like bridge to the compact-group size study of Coenda et al. (2012). All sizes are required to lie between 0.1 and 50 kpc after re-conversion; Sérsic fits pegged at the GIM2D bounds ($n_g \leq 0.55$ or $n_g \geq 7.9$; 60 galaxies) are excluded from the primary sample — per sample, 5 of 197 Simard-matched CG_4 members (2.5%), against 22/1939, 31/2061, and 2/199 in $\text{Control}_{4\text{B}}$, $\text{Control}_{4\text{C}}$, and RG_4 (1.0–1.5%): the cut is mildly differential in the direction expected if pegged fits preferentially affect concentrated early types, so removing them is conservative with respect to the CG_4 early-type excess — and Petrosian entries must satisfy $R_{50,r} > 0$, $R_{90,r} > R_{50,r}$, and a finite positive uncertainty. The completeness of both measures, and its difference between CG_4 and the controls, is audited in Sect. 3.8.

2.4. Statistical design

Descriptive baseline comparisons use two-sided Fisher exact tests for ordinary 2×2 categorical tables, and rank-sum or bootstrap comparisons for distributions and median differences. Spearman and Pearson coefficients are used only as diagnostics of monotonic or linear trends; small coefficients are interpreted as weak even when precisely estimated.

We organize the inference around a single astrophysical question: after accounting for stellar mass, redshift, galaxy rank, and group-scale differences, do compact groups retain a distinct galaxy population? Separate $\text{Control}_{4\text{B}}$, $\text{Control}_{4\text{C}}$, and RG_4 comparisons reveal how the answer changes from luminous ordinary-group members to projected cores and true four-member groups. Galaxy-level models establish the adjusted population contrast; group-level matching tests whether whole quartets contain a different satellite mix; and within-host models ask whether compact-subconfiguration members differ from their non-compact co-members. Binary complements are represented once: elliptical versus spiral among usable Galaxy Zoo classifications, and quenched versus star-forming among galaxies with valid sSFR.

Throughout, the independence unit for inferential contrasts among grouped catalogues is the *physical* group: standard errors are clustered by Lim group (unifying the overlapping control definitions, with CG_4 systems assigned to their host Lim group), and resampling of grouped contrasts is blocked by physical group or matched group/pair block, since the four galaxies of one group are not independent observations. Pooled models across the three controls are reported only as secondary summaries, after deduplicating the overlapping control samples to one row per physical galaxy (Appendix F). Paired sign-flip permutation p-values use add-one tail counts, and bootstrap sign-crossing p-values use $2 \min[(k_{\leq 0} + 1)/(B + 1), (k_{\geq 0} + 1)/(B + 1)]$, capped at unity; displayed p-values are floored at 10^{-4} . Purely descriptive reference-sample diagnostics report their own unit in the relevant appendix.

Throughout, the velocity dispersion σ_v that enters models and matching is the quartet-level gapper estimate (Wainer & Thissen 1976) on the four selected members’ redshifts, converted as $c \sigma_z / (1 + \bar{z})$ — for *all* samples, including $\text{Control}_{4\text{B}}$ and $\text{Control}_{4\text{C}}$ quartets embedded in richer hosts, whose σ_v therefore describes the selected quartet rather than the host halo (this is how the spurious Lim

3688 quartets of Sect. 2.1 reach $\approx 2000 \text{ km s}^{-1}$). With four tracers this estimator is intrinsically noisy: in a Gaussian Monte Carlo of 10^5 quartets, $\sigma_{\text{gapper}}/\sigma_{\text{true}}$ has median 0.96 with a 16–84% range of 0.57–1.42. Candidate covariates enter an adjusted model when non-missing for at least 65% of that model’s frame (matching variables require 70%); in practice σ_v is complete for every quartet and always enters, the group-luminosity, stellar-mass, redshift, and rank covariates are likewise complete, and no harmonised group-mass covariate is available (Appendix F.3). Because σ_v is noisy, all analyses that use it are repeated without it; the scientific conclusions are unchanged (Sects. 3.2, 3.3).

The three control samples are pre-specified, physically distinct comparison populations and are therefore reported as separate contrasts rather than treated as repeated tests of one common null hypothesis. Elliptical/spiral and quenched/star-forming are each represented by one binary outcome. We report ordinary two-sided p-values for the planned morphology, star-formation, matching, host, and size analyses, together with effect sizes and confidence intervals. Multiplicity control is reserved for broad exploratory batteries: Holm adjustment for the colour and measurement-availability screens and for the magnitude-gap batteries, and Benjamini–Hochberg false-discovery-rate control for the larger dominance and morphology–environment screens. The paired sign-flip permutation p-value is used for group-level matches; bootstrap intervals quantify effect-size uncertainty. Colours, AGN-like fractions, H α equivalent widths, magnitude gaps, crossing-time trends, projected phase space, and the pairwise tidal index remain secondary diagnostics.

3. Results

3.1. Raw population contrasts

The sSFR status and Galaxy Zoo morphology mix of each sample are shown in Tables 1 and 2. Among galaxies with a usable elliptical or spiral vote classification, a single two-sided Fisher test of the E/Sp mix gives $p = 6.8 \times 10^{-4}$ against Control_{4B}, $p = 1.9 \times 10^{-3}$ against Control_{4C}, and $p < 10^{-4}$ against RG₄. Thus CG₄ galaxies show a clear early-type-like shift, with more elliptical-vote and fewer spiral-vote classifications. The usable comparison contains 210 CG₄ galaxies; 38 rows with uncertain or missing Galaxy Zoo classifications are excluded rather than assigned to either outcome. Among usable BGG classifications, the elliptical fraction is 67.9% in CG₄ and 44.7% in RG₄; the Fisher p-values are 1.00, 1.00, and 0.03 against Control_{4B}, Control_{4C}, and RG₄. Restricted to star-forming galaxies with usable E/Sp classifications, the mix is compatible with every control (two-sided Fisher exact test: $p = 0.332$, $p = 0.468$, and $p = 0.226$, same control order). The raw morphology excess is therefore not carried by star-forming ellipticals.

The star-formation contrast is weaker and rank-dependent. In the raw all-galaxy comparison, Fisher’s exact test gives star-forming-fraction p-values of $p = 0.481$ versus the Control_{4B} sample, $p = 0.262$ versus Control_{4C}, and $p = 3 \times 10^{-4}$ versus RG₄. For satellites, the corresponding p-values obtained with two-sided Fisher exact test are $p = 0.302$ against Control_{4B}, $p = 0.114$ against Control_{4C} and $p = 2.3 \times 10^{-4}$ against RG₄; for BGGs, they are $p = 0.578$, $p = 0.577$, and $p = 0.37$ (same control order). The satellite deficit is clearest against RG₄. We treat all

Table 1. Number of galaxies in each sSFR class for each sample. Galaxies without a valid sSFR estimate (‘No sSFR’) are excluded from the classification: their percentages are quoted with respect to all galaxies of the (sub)sample, while the quenched and star-forming percentages are quoted with respect to classified galaxies only. The SDSS non-AGN reference extends to substantially fainter absolute magnitudes than the CG₄ sample and is shown for orientation only; it is not a control sample and enters no test in this paper.

Sample	No sSFR	Quenched	Star-forming
CG ₄	18 (7%)	146 (63%)	84 (37%)
BGG	6 (10%)	45 (80%)	11 (20%)
Satellites	12 (6%)	101 (58%)	73 (42%)
Control _{4B}	180 (6%)	1593 (61%)	1019 (39%)
BGG	58 (8%)	533 (83%)	107 (17%)
Satellites	122 (6%)	1060 (54%)	912 (46%)
Control _{4C}	191 (7%)	1560 (60%)	1061 (40%)
BGG	59 (8%)	537 (83%)	107 (17%)
Satellites	132 (6%)	1023 (52%)	954 (48%)
RG ₄	5 (2%)	101 (46%)	118 (54%)
BGG	3 (5%)	38 (72%)	15 (28%)
Satellites	2 (1%)	63 (38%)	103 (62%)
SDSS ref.	634 (1%)	7705 (15%)	44192 (85%)

Table 2. Number of galaxies in each Galaxy Zoo vote-fraction morphology proxy for each sample. Percentages are quoted with respect to each sample’s final row count; the inferential E/Sp comparison instead conditions on the usable E/Sp subset. The No GZ column counts rows without finite Galaxy Zoo vote fractions after merging and is separated from the finite-vote uncertain class. As in Table 1, the SDSS non-AGN reference extends to substantially fainter absolute magnitudes than the CG₄ sample and is not a control sample.

Sample	Ellip.	Spiral	Unc.	No GZ
CG ₄	124 (50%)	86 (35%)	7 (3%)	31 (12%)
Control _{4B}	1106 (40%)	1266 (45%)	116 (4%)	304 (11%)
Control _{4C}	1118 (40%)	1229 (44%)	144 (5%)	321 (11%)
RG ₄	60 (27%)	129 (58%)	15 (7%)	20 (9%)
SDSS ref.	10091 (19%)	34715 (66%)	7725 (15%)	0 (0%)

raw tests as descriptive, because star-formation class and morphology are entangled with stellar mass and luminosity; this motivates the adjusted analyses of Sect. 3.2.

For the star-forming subset, we fit the star-forming main sequence with polynomials of order 1–4, adopting order 2 as the lowest-order model with the smallest five-fold cross-validated RMS scatter (0.305 dex; Appendix D). The median CG₄-minus-control residual offsets about this relation are small, control-dependent, and change in strength after matching (Appendix D), so we treat main-sequence residuals as a secondary diagnostic rather than as evidence for a robust compact-group star-formation offset.

3.2. Morphology and star formation at fixed galaxy properties

The three control samples answer different questions, so we fit CG₄ against each control *separately*: Control_{4B} represents the luminous population of eligible ordinary groups, Control_{4C} their BGG-centred projected cores, and RG₄ true four-member ordinary groups. Each contrast uses the same covariate set (stellar mass, redshift, rank, group luminosity, and velocity dispersion) and clusters standard errors by physical group. We model one outcome from each binary pair: elliptical rather than spiral morphology among usable Galaxy Zoo classifications, and quenched rather than star-forming status among valid sSFR measurements.

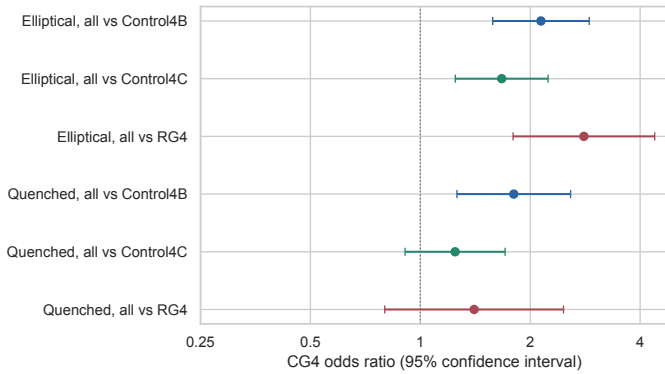


Fig. 1. Adjusted CG₄ odds ratios from the three separate per-control contrasts (colour-coded by control sample) for the all-member elliptical and quenched outcomes. Bars show 95% confidence intervals; the vertical line marks equal odds.

Against Control_{4B}, the elliptical odds ratio is 2.14 (95% CI [1.58, 2.90]; $p < 10^{-4}$), and the quenched odds ratio is 1.80 (95% CI [1.26, 2.58]; $p = 0.001$).

Against Control_{4C}, the elliptical odds ratio is 1.67 (95% CI [1.25, 2.24]; $p = 5.9 \times 10^{-4}$), and the quenched odds ratio is 1.25 (95% CI [0.91, 1.71]; $p = 0.173$).

Against RG₄, the elliptical odds ratio is 2.81 (95% CI [1.79, 4.38]; $p < 10^{-4}$), and the quenched odds ratio is 1.40 (95% CI [0.80, 2.47]; $p = 0.238$).

The gradation across the contrasts is itself informative, but it is not by itself a statistical test of differences between the three effect estimates. The adjusted individual-galaxy morphology excess is strong against the luminosity-selected cores of richer groups and against true four-member groups, and smallest against the BGG-centred *projected* cores of ordinary groups (Control_{4C}), the control definition that most closely mimics a compact projected configuration. This ordering is consistent with a compactness or projected-density sequence — the excess is largest against the two ~440 kpc environments (Control_{4B}, RG₄) and smallest against the ~293 kpc Control_{4C} cores (Sect. 2.1) — and the group-level satellite-composition contrast below shows the same pattern, but it does not prove a causal compactness sequence. It motivates the local projected-density diagnostic of Sect. 3.6 (Fig. 1).

The direct mass-binned view in Fig. 2 shows the same physical distinction without another model: through the well-populated intermediate-mass bins, CG₄ has higher median elliptical-vote and lower spiral-vote fractions, while Control_{4C} is generally the closest comparison. The satellite quenched-fraction curves cross, and the BGG curves are similar. Thus the visual trends support the morphology result but do not reveal an additional CG₄-specific quenching pattern.

As a compact secondary summary, a pooled model over the deduplicated control catalogues (one row per physical galaxy, cluster-robust by physical Lim group; Appendix F) gives the same qualitative picture: the pooled elliptical odds ratio is 1.77 ($p = 1.4 \times 10^{-4}$), and the pooled quenched odds ratio is 1.35 ($p = 0.068$). The adjusted morphology signal is primarily carried by satellites: their elliptical odds ratio is 2.08 (95% CI [1.51, 2.88]; $p < 10^{-4}$), whereas the BGG estimate is 0.78 (95% CI [0.39, 1.53]; $p = 0.464$) (Fig. 3).

As announced in Sect. 2.4, the per-control and pooled models were refit without the noisy σ_v covariate on iden-

tical samples and with identical clustering. The elliptical odds ratios move by less than their standard errors and the scientific interpretation is unchanged.

A host-BGG alignment audit supports the satellite interpretation: 56 of 62 non-split CG₄ systems have the same quartet and Lim-host BGG objid. The 6 BGG-misaligned systems are Predominant systems; separately, the 6 Isolated systems are trivial quartet=host equality cases. All 186 quartet-rank CG₄ satellites are Lim-host satellites. Restricting CG₄ to the 56 BGG-aligned systems leaves the three satellite elliptical contrasts stable; against Control_{4C} the aligned odds ratio is 1.98.

3.3. Matched comparisons

Satellite composition of matched quartets. Because the four galaxies of a compact group are not independent, we treat each quartet as the statistical unit: for each group we count its elliptical-vote satellites ($N_{\text{ell,sat}} \in \{0, 1, 2, 3\}$) and match CG₄ groups one-to-one to controls on redshift, BGG stellar mass, total quartet luminosity, and velocity dispersion (propensity caliper 0.2 SD of the logit). We repeat the match separately for each control type and report paired sign-flip permutation p-values with group-bootstrap confidence intervals. A deduplicated pooled match is retained only as a compact secondary summary.

The pooled control match covers 54 CG₄ groups and 633 control groups, yielding 54 matched pairs, drawn predominantly from Control_{4B} (46 Control_{4B}, 6 RG₄, 2 Control_{4C}, reflecting the pool composition after deduplication). This pooled match is a secondary summary because the control catalogues overlap. Its elliptical-vote satellite fraction difference is $\Delta = 0.167$ (95% bootstrap interval [0.040, 0.290]; permutation $p = 0.016$, bootstrap $p = 0.010$; $B = 9999$).

The per-control matches (Table 3) give a consistent ordering: the CG₄-minus-control elliptical-satellite fraction is positive against every control, clearest against Control_{4B} and RG₄, and weakest against Control_{4C}. The physically most stringent Control_{4C} match yields 54 pairs with $\Delta = 0.114$ (95% CI [-0.015, 0.238]; permutation $p = 0.080$). The pairs are already close to balanced in mean satellite stellar mass (paired differences of 0.00, 0.03, and -0.01 dex against Control_{4B}, Control_{4C}, and RG₄). Adding satellite mass to the match preserves the same hierarchy: $\Delta = 0.198$, 0.086, and 0.256, respectively. Within the common $\Delta m \leq 3$ window, distance-ranked Control_{4C} satellites sample their parents' eligible satellite distribution much more democratically than brightness-ranked Control_{4B} satellites (KS $D = 0.04$ and 0.30), which helps explain why Control_{4C} is the closest ordinary-group analogue.

Because σ_v is a noisy matching covariate (Sect. 2.4), all matches were repeated without it: the galaxy-level match retains the same 234 CG₄ galaxies with an unchanged conclusion pattern, and the per-control group-level differences become $\Delta = 0.284$, 0.102, and 0.253 against Control_{4B}, Control_{4C}, and RG₄ (permutation $p = 0.0001$, 0.056, 0.0016) — the same qualitative picture, with Control_{4C} again the weakest contrast.

Matched individual galaxies. We matched 234 of the 248 available CG₄ galaxies one-to-one to 234 ordinary-group galaxies without replacement, requiring common propensity-score support, a 0.2-standard-deviation caliper,

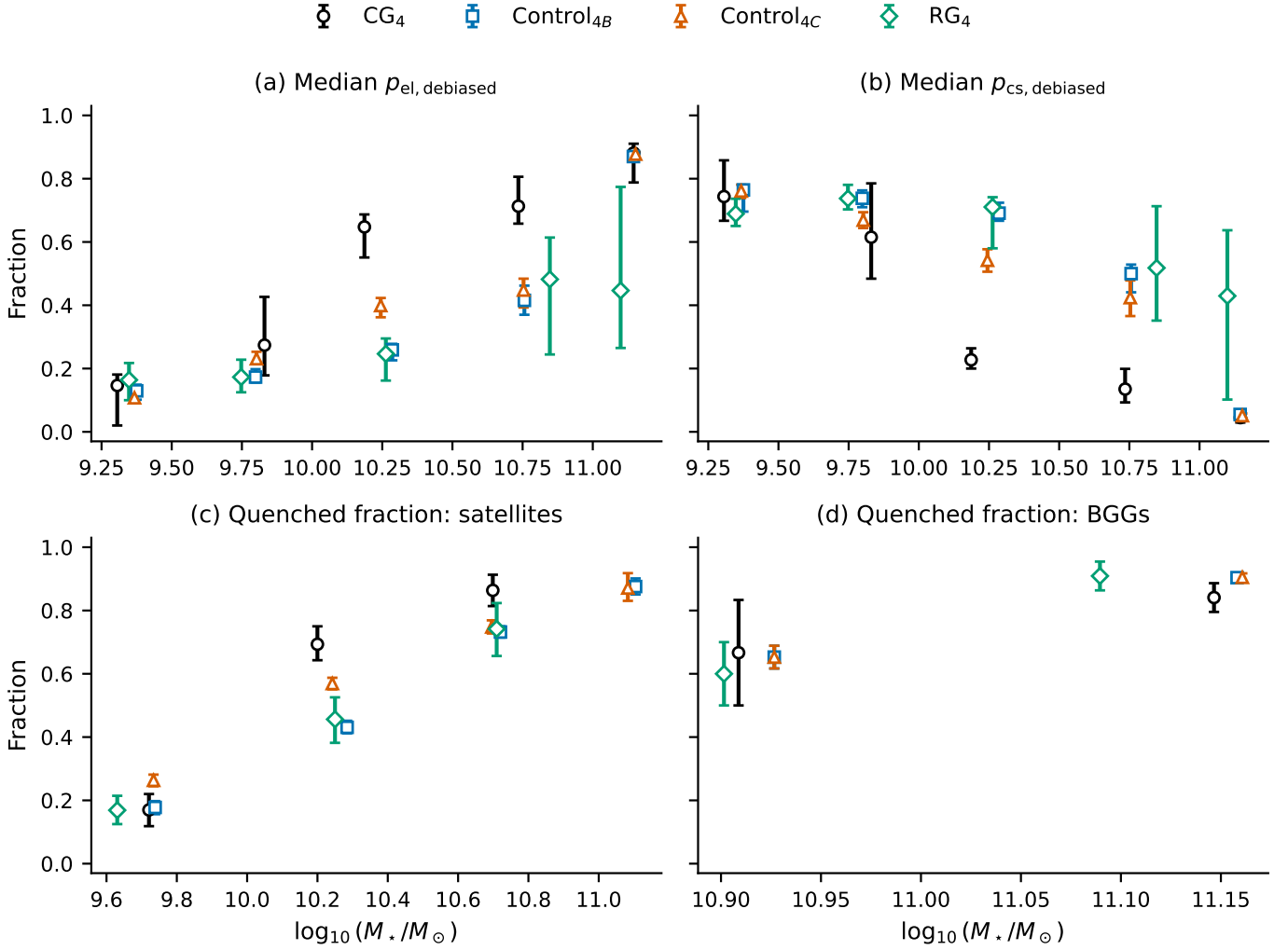


Fig. 2. Galaxy Zoo vote fractions and quenched fractions versus stellar mass. Top: median debiased elliptical and spiral vote fractions for all galaxies. Bottom: quenched fractions among valid-sSFR satellites and BGGs. All samples use the same bins within each row: edges 7.0, 9.5, 10.0, 10.5, 11.0, 12.5 for morphology and 7.0, 10.0, 10.5, 11.0, 12.5 for quenching. Each estimate is plotted at the median stellar mass of the galaxies contributing to it, rather than at the geometric bin centre; points are not connected because the displayed bins have unequal widths. Bars are 16th–84th percentile physical-group-blocked bootstrap intervals; bins with fewer than five galaxies or three groups are omitted.

Table 3. Per-control group-level matched satellite-composition contrasts. Δ is the matched mean difference in elliptical-vote satellite fraction (CG_4 minus control). The paired sign-flip permutation p-value is the inferential test; the bootstrap p-value is an add-one sign-crossing diagnostic, with $B = 9999$ resamples. The pooled row is secondary. Pair counts: 54 CG_4 groups have complete matching covariates and classified satellites; against Control_{4B} and Control_{4C} (630 and 632 available control groups) every CG_4 group finds a within-caliper partner, whereas the RG_4 pool retains 28 pairs on common support.

Control	Pairs	Δ	95% CI	$p_{\text{perm}} / p_{\text{boot}}$
Control_{4B}	54	0.160	[0.034, 0.284]	0.016 / 0.012
Control_{4C}	54	0.114	[−0.015, 0.238]	0.080 / 0.082
RG_4	28	0.196	[0.036, 0.351]	0.026 / 0.019
Pooled	54	0.167	[0.040, 0.290]	0.016 / 0.010

0.08. Confidence intervals and p-values come from a bootstrap blocked by the treated galaxy’s compact group; pool provenance and balance are documented in Appendix F.

The matched CG_4 -minus-control difference in elliptical fraction is 0.139, with 95% bootstrap interval [0.037, 0.240] and $p = 0.007$. For satellite pairs alone, the difference rises to 0.192 (95% CI [0.067, 0.313]; $p = 0.004$; 130 pairs), reinforcing the conclusion that satellites carry the morphology shift. The matched quenched-fraction difference is 0.031 (95% CI [−0.056, 0.122]; $p = 0.526$), which does not establish an independent quenching excess. Among matched star-forming galaxies, the residual-sSFR difference is −0.147 ([−0.311, 0.008]; $p = 0.064$).

3.4. Compact-group members versus other members of the same host group

For the Embedded and Predominant CG_4 systems, whose quartets live inside richer Lim parent groups, the sharpest available test of “compact subconfiguration versus shared

and exact rank strata. The propensity model uses $\log M_*$, redshift, rank, $\log L_{\text{group}}$, and σ_v . The maximum absolute covariate standardized mean difference falls from 0.37 to

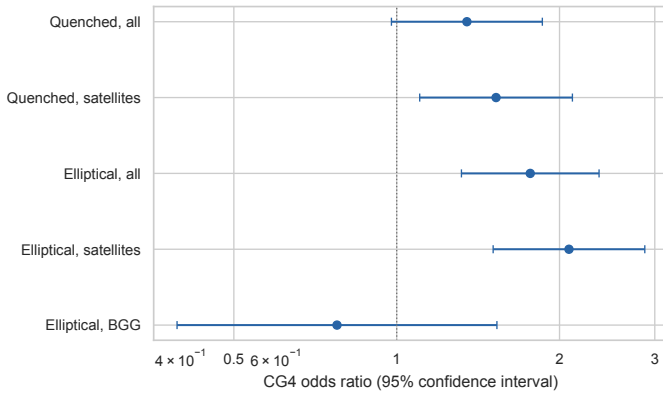


Fig. 3. Adjusted CG_4 odds ratios for the unique star-formation and morphology outcomes in the pooled secondary models, including all-galaxy, satellite-only, and BGG-only rows. Bars show 95% confidence intervals; the vertical line marks equal odds.

host environment” compares the CG_4 member galaxies with the *non- CG_4 members of the same host group*. The primary estimator is conditional logistic regression stratified by host group: the stratum-level intercept is conditioned out in the partial likelihood, so every host-level confounder shared by all members of the same host (halo mass, richness, redshift, large-scale environment) is removed by design. As a robust-

ness check we also fit a binomial GLM with host-group fixed-effect dummies and cluster-robust standard errors by host (FE-GLM). The design starts from 54 host groups (958 member galaxies, of which 224 belong to a CG_4); the models include stellar mass, host luminosity rank, and projected host-centric radius as galaxy-level covariates.

For the elliptical-vote outcome, the complete-case conditional-logit sample contains 791 galaxies in 53 host strata; all complete-case host strata are informative. The elliptical-vote odds ratio for CG_4 membership is 1.47 (95% CI [0.92, 2.35]; $p = 0.108$), and the quenched odds ratio is 0.75 (95% CI [0.45, 1.25]; $p = 0.269$; 915 complete cases).

The FE-GLM sensitivity fit gives OR 1.57 (95% CI [1.02, 2.43]; $p = 0.041$). We prefer conditional logit because its conditional likelihood eliminates the nuisance host intercepts rather than estimating many indicators from small strata, where incidental-parameter concerns arise; the FE-GLM remains a sensitivity analysis, and its different uncertainty alone does not establish bias or artefact in either fit. At fixed stellar mass, rank, and host-centric radius, the preferred within-host estimator does not show a clear additional morphology excess for compact-subconfiguration membership. Its confidence interval still permits moderate effects. Together with the smaller CG_4 –Control_{4C} contrast, this points to dense projected-core location and host environment rather than a categorical effect of the compact-group label alone.

The radius covariate defines the question: conditioning on projected host-centric radius asks whether CG_4 members differ *beyond their central location*. Without that term, the conditional-logit elliptical odds ratio rises to 1.90 (95% CI [1.23, 2.95]; $p = 0.004$), with a similar direction from the FE-GLM, while the quenched outcome remains weak (radius-free conditional-logit OR 1.22; $p = 0.412$). This is the pattern expected if CG_4 galaxies preferentially occupy the early-type-rich central regions of their hosts.

Table 4. Robustness of the CG_4 morphology contrast. Odds ratios (OR) are adjusted CG_4 coefficients; the continuous row is the adjusted offset in the debiased elliptical vote fraction p_E . P-values are unadjusted.

Test	Result
55-arcsec neighbour exclusion	all-row E: 52.5% / 40.1% (CG_4 / Control _{4C}); E/S Fisher $p = 0.002$
Add close-neighbour flag to model	CG_4 OR 1.76 [1.30, 2.39]; $p = 2.6 \times 10^{-4}$
DS18 $TType \leq 0$	OR 2.11 / 1.49 / 2.22 (Control _{4B} / Control _{4C} / RG ₄); all $p \leq 0.017$
Threshold sweep 0.4–0.8	OR 1.73–2.19
Continuous p_E offset	0.089 [0.047, 0.131]; $p < 10^{-4}$
Sérsic $n_g > 2.5$	OR 1.60; $p = 0.002$

3.5. Morphology robustness

Because Galaxy Zoo morphology is image-based and the adopted vote-fraction cut is one of several defensible choices, we subject the morphology contrast to a battery of robustness tests, summarized in Table 4.

After excluding galaxies whose nearest projected neighbour within the catalogue group lies within 55 arcsec (retaining 200 CG_4 and 2534 Control_{4C} galaxies), the CG_4 elliptical fraction *rises* relative to the raw full-sample value in Table 2, rather than falling, and the CG_4 –Control_{4C} contrast remains clear. The quoted 52.5% and 40.1% use all retained rows (including U and NoGZ), whereas the Fisher test uses only E and S. In the joint model, the close-neighbour flag has OR 1.98 ($p = 6.1 \times 10^{-4}$), while the CG_4 coefficient conditional on that flag is OR 1.76 ($p = 2.6 \times 10^{-4}$). The former compares close with non-close galaxies; the latter compares CG_4 with controls at fixed flag status. These distinct conditional quantities, and the unequal excluded shares, explain why neither odds ratio contradicts the rising raw CG_4 E fraction after exclusion. The checks do not prove immunity to crowding, but rule out the closest-pair artefact as the sole origin of the result.

We also refit the per-control adjusted models after the same 55-arcsec exclusion, which removes 19% of CG_4 members against 6–10% of the controls. The all-member elliptical odds ratio moves from 2.14 to 2.16 against Control_{4B}, from 1.67 to 1.71 against Control_{4C}, and from 2.81 to 2.71 against RG₄. The adjusted excess is therefore not carried by the crowded closest pairs.

Across classification thresholds of 0.4–0.8 on the deduplicated pooled sample, the adjusted CG_4 elliptical odds ratio varies only mildly and never changes sign (Table 4; the unclassified fraction grows from 13% to 46% over the sweep). A threshold-free check models the debiased elliptical vote fraction p_E itself, and a structural cross-check independent of Galaxy Zoo uses the Simard et al. (2011) pure-Sérsic index (early type: $n_g > 2.5$; 75% coverage); both confirm the direction of the contrast, the latter subject to the differential Simard completeness discussed in Sect. 3.8.

Using the same per-control covariates, standardization, and physical-group clustering, an independent DS18 $TType \leq 0$ outcome gives positive CG_4 coefficients in all three contrasts (Table 4; $n = 1822, 1913, 324$). Support is broad except that 83% of CG_4 values lie within the RG₄ group-luminosity range, limiting that smallest comparison.

3.6. Pairwise tidal index

The pairwise analysis reconstructs projected separations and velocity differences for 6076 catalogue rows and refits adjusted models with and without the pairwise tidal index $T_i = \sum_j M_{\star,j}/R_{ij}^3$ (in $M_\odot \text{ kpc}^{-3}$). The sum runs over each entry's *selected quartet co-members only* — not the full Lim host membership, a restriction shown to be innocuous in Appendix F.3 — with no velocity window; R_{ij} converts the great-circle separation with the angular-diameter distance at the quartet's median redshift (proper kpc); co-members with unmeasured stellar mass contribute zero, making T_i a lower bound in those groups, and quartets with any member lacking coordinates or redshift are dropped whole. These exploratory models adjust for stellar mass and satellite/BGG status only, pool the three control catalogues *without* objid deduplication (unlike the pooled models of Appendix F), and cluster standard errors by physical group. The two model variants estimate *different quantities*. The model without the index estimates the *total* association between compact-group membership and the outcome at fixed stellar mass and satellite status; the model with the index estimates the association *at fixed projected local density*. Because the index is built from R^{-3} -weighted projected separations, it is mathematically entangled with the compact-group selection itself (compact groups are selected to have small projected separations), so conditioning on it partially conditions on the exposure. A reduction of the CG₄ coefficient when the index is added is thus *conditional attenuation*, not mediation: it does not show that local density accounts for the morphology contrast, and it cannot, because the causal ordering of selection, density, and morphology is not identified by projected snapshots. The baseline models in this subsection are refit on the pairwise-analysis complete-case sample, so their coefficients need not be numerically identical to the corresponding coefficients in Sect. 3.2 or Appendix F; because R_{ij} is projected, the term is a local projected-density and immediate-neighbour proxy rather than a direct tidal-history measurement.

For elliptical morphology, the baseline CG₄ odds ratio is 1.95 ($p < 10^{-4}$), whereas the model including the tidal index gives a CG₄ odds ratio of 1.39 ($p = 0.031$); the tidal-index term itself has odds ratio 1.42 per 1.07-dex standard deviation of $\log_{10} T_i$ ($p < 10^{-4}$). The projected tidal-density proxy therefore *attenuates* the CG₄ coefficient; given the selection entanglement described above, this conditional attenuation must not be read as the index accounting for the effect, and the result remains an exploratory projected-pair diagnostic.

For quenched status, the baseline CG₄ odds ratio is 1.48 ($p = 0.014$), while the model including the tidal index gives 1.02 ($p = 0.928$), with a tidal-index term of odds ratio 1.51 ($p < 10^{-4}$).

Common-support diagnostics quantify how far the index can be conditioned on. The CG₄ and control $\log T_i$ distributions have an overlap coefficient of 0.54, but their medians differ by 1.1 dex (per control: +1.34 dex versus Control_{4B}, +0.81 versus Control_{4C}, and +1.22 versus RG₄) and 15% of CG₄ galaxies lie above the pooled-control 95th percentile of $\log T_i$; collinearity in the adjusted design is nevertheless mild (point-biserial $r = 0.18$; variance-inflation factor of the CG₄ indicator 1.03). Sensitivity refits reflect that thin far tail: the elliptical CG₄ odds ratio at fixed T_i is 1.39 in the linear fit above, 1.40 with a natural cubic spline in $\log T_i$, 1.27 after deduplicating the control pool

by objid ($p = 0.12$), and 1.22 ($p = 0.20$) when trimmed to the common-support window of $\log T_i$. The residual CG₄ coefficient at fixed projected density is therefore not stable enough to bear physical weight, reinforcing the conditional-attenuation reading above.

3.7. Secondary outcomes

Star formation. The per-control quenched odds ratios vary with the comparison sample (Sect. 3.2), the matched quenched fraction shows no clear offset (Sect. 3.3), and main-sequence residuals provide only a secondary check. The largest per-control main-sequence offset is against Control_{4C} ($\Delta = 0.058$, 68% interval 0.039–0.103; $p = 0.030$; Appendix D), whereas the matched residual-sSFR difference is -0.147 ($p = 0.064$). Because the two comparisons account for different galaxy properties, they do not support a stable compact-group main-sequence displacement.

The interpretation does not rely on the exact GMM division of the sSFR plane: repeating the key satellite comparison with a fixed threshold, $\log_{10}(\text{sSFR}/\text{yr}^{-1}) = -11.0$, gives CG₄ and RG₄ satellite star-forming fractions of 41.4% and 60.2%, respectively, a difference of -18.8 percentage points ($p = 5.3 \times 10^{-4}$). The qualitative hierarchy is unchanged: the morphology result is independent of the sSFR threshold, whereas the quenched/star-forming contrast remains more sensitive to modelling choices.

H α emission. Our retrieved catalogues do not include the age-sensitive indices D_n4000 and $\text{H}\delta_A$ (available in the MPA-JHU `galSpecIndx` product), so we defer a post-starburst classification to follow-up work and perform only a limited H α -equivalent-width comparison, using the SDSS convention in which negative equivalent width denotes emission and classifying strong H α emission by $\text{H}\alpha \text{EW} \leq -3 \text{ \AA}$.

The strong-emission fraction is 32.3% in CG₄, 41.0% in Control_{4B}, 43.3% in Control_{4C}, 56.9% in RG₄.

Relative to Control_{4B}, the CG₄ strong-emission fraction differs by -0.087 ($p = 0.012$).

Relative to Control_{4C}, the CG₄ strong-emission fraction differs by -0.110 ($p = 0.002$).

Relative to RG₄, the CG₄ strong-emission fraction differs by -0.246 ($p < 10^{-4}$).

These strong-H α fractions are reported as *descriptive* diagnostics only: they are not adjusted for the samples' different morphology mixes, and since LINER-like emission is common in early-type galaxies, a morphology-blind emission fraction partially restates the morphology contrast itself. They indicate a lower strong-emission fraction in CG₄ than in each control, but provide neither a direct post-starburst or recently-quenched classification nor an independent star-formation test.

Emission-line and AGN-like populations. The emission-line analysis classifies BPT star-forming, composite, and AGN-like populations for galaxies with the required diagnostic line ratios: composite galaxies lie between the Kauffmann et al. (2003a) and Kewley et al. (2001) demarcations in the [NII] BPT plane, and AGN-like galaxies lie above the Kewley et al. (2001) line; galaxies with a pre-existing AGN flag but no BPT line ratios are also counted as

AGN-like, and BPT-unclassified galaxies are excluded from the AGN-like fraction denominator rather than treated as confirmed non-AGN. Among BPT-classified CG₄ galaxies, the AGN-like fraction is 38.0%.

After mass and environment adjustment, the CG₄ AGN-like odds ratio is 1.17 ($p = 0.364$): AGN-like activity does not provide a robust independent compact-group discriminator.

Magnitude gaps and fossil-like assembly. We define $\Delta m_{12} = M_{r,2} - M_{r,1}$ and $\Delta m_{14} = M_{r,4} - M_{r,1}$, so that a more dominant BGG has a positive, larger gap; large gaps are the classic optical signature of fossil-like assembly (Ponman et al. 1994; Jones et al. 2003; Dariush et al. 2010).

Under the magnitude-gap adjustment described in Sect. 2.4, larger gaps appear only against Control_{4B}: for Δm_{12} , the CG₄ and Control_{4B} medians are 1.17 and 0.85 mag (adjusted $p < 10^{-4}$), and for Δm_{14} they are 2.56 and 1.90 mag (adjusted $p < 10^{-4}$).

The corresponding Control_{4C} and RG₄ differences are weak for both gap definitions.

A deduplicated pooled sensitivity gives control medians of 0.86 mag for Δm_{12} and 1.91 mag for Δm_{14} (adjusted $p < 10^{-4}$ and $p < 10^{-4}$). Because the deduplicated control side is dominated by Control_{4B}-selected systems, it is not evidence for a control-robust gap excess.

The compact-group morphology signal is therefore not accompanied by a comparably robust excess in these simple fossil-like magnitude-gap indicators.

In the separate 14-test gap-correlation screen, the relation between Δm_{12} and quenched satellite fraction is $\rho_s = -0.18$ (adjusted $p < 10^{-4}$).

Projected phase space. We ask whether the CG₄–RG₄ satellite contrasts vary with projected position relative to the BGG. The analysis uses 174 CG₄ and 166 RG₄ satellites, excludes rank-1 galaxies, ranks satellites by projected BGG-centric distance, and, when available, uses $|\Delta v_{\text{BGG}}|/\sigma_v$ as a projected phase-space diagnostic; such coordinates are statistical proxies for infall state or quenching time, not orbital reconstructions (Mahajan et al. 2011; Oman & Hudson 2016; Wetzel et al. 2013).

In the innermost satellite-distance bin, the quenched fractions are 68.3% for CG₄ and 48.2% for RG₄, a difference of 20.1 percentage points; the corresponding early-type fractions are 64.7% and 35.6% (difference 29.1 points).

The quenched-fraction contrast is similar between the inner and outer distance ranks, so the data do not isolate the CG₄ offset to the immediate BGG environment.

After adjusting for stellar mass, redshift, distance bin, and available group-scale controls, the quenched-status model gives a CG₄ odds ratio of 1.18 (95% CI 0.43–3.24; $p = 0.742$), and the analogous early-type model gives 2.42 ($p = 0.078$).

A nearest-neighbour mass–redshift matched check on 126 CG₄–RG₄ satellite pairs gives matched quenched- and early-type-fraction differences of 9.5 and 18.7 percentage points, respectively.

In compact groups the member counts are small, velocity dispersions are noisy, and projection mixes recently accreted and long-resident satellites, so we treat these bins as an observational diagnostic of where the contrast appears, not as a dynamical clock.

Optical colours. We derive the $(u - r)$, $(u - g)$, $(g - r)$ and $(r - i)$ colours from the matched SDSS photometry and model each colour jointly with $\log(M_\star/M_\odot)$ and redshift. The colour subset is not representative of the full catalogues – matched galaxies are systematically less massive, more actively star-forming, and more frequently satellites than unmatched galaxies – and the coefficients depend on the control sample. It therefore provides no robust global blue or red offset for CG₄ satellites (Appendix E.6).

Group-scale diagnostics. A broader set of environmental diagnostics is reported in Appendix E: sample-specific monotonic sSFR–distance trends (Appendix E.1); dominance splits whose clearest separation is in group luminosity rather than in any CG₄-specific property; morphology tests against the Zheng–Shen class, Δm_{12} , and the BGG luminosity fraction, none of which survives false-discovery-rate control; and a crossing-time diagnostic that primarily tracks virial mass and mass-to-light ratio.

3.8. Galaxy sizes at fixed stellar mass

We now test whether CG₄ galaxies differ in size at fixed stellar mass, using the seeing-corrected Simard half-light radii as the primary measure and the Petrosian radii as the model-independent cross-check (Sect. 2.3). The Simard measure resolves 77.4% of the CG₄ galaxies and 71.1% of the pooled control galaxies, while the Petrosian radius is essentially complete in every sample (Appendix Fig. F.2). Because the CG₄–control difference in Simard completeness exceeds five percentage points, every Simard-based coefficient below carries a differential-completeness caveat and is read together with the Petrosian rerun, which does not share this incompleteness.

For satellites, the CG₄ size offset at fixed stellar mass, redshift, and available group-scale covariates is -0.010 dex (-2.3%), with 95% confidence interval $[-0.039, 0.019]$ and $p = 0.507$, over 2575 satellites in 754 physical groups.

The all-galaxy model gives -0.006 dex (95% CI $[-0.032, 0.020]$; $p = 0.653$), and the BGG-only model gives 0.019 dex (95% CI $[-0.037, 0.075]$; $p = 0.503$).

The pooled adjusted models therefore show no meaningful fixed-mass size displacement. The per-control estimates are -0.035 dex against Control_{4B}, 0.000 dex against Control_{4C}, -0.006 dex against RG₄ (Appendix Fig. G.1). The isolated negative Control_{4B} estimate is not repeated for the two physically different controls and is not promoted to a general compact-group size effect.

The one-to-one matched comparison uses the same 234 propensity pairs as the galaxy-property analysis. For the 166 pairs with valid Simard sizes, the mean CG₄-minus-control difference is -0.019 dex (95% bootstrap interval $[-0.064, 0.026]$; $p = 0.400$). The Petrosian difference is -0.033 dex (95% CI $[-0.065, -0.003]$; $p = 0.030$), and the concentration difference is -0.008 ($p = 0.904$). The Petrosian-only offset is small, depends on the size definition and matched control composition, and does not overturn the near-null seeing-corrected result.

The Petrosian rerun, with seeing as an additional standardized covariate, gives a satellite coefficient of -0.019 dex ($p = 0.111$) and an all-galaxy coefficient of -0.012 dex ($p = 0.206$).

The concentration index $C = R_{90,r}/R_{50,r}$ provides a model-independent structural proxy: the adjusted satellite coefficient is 0.068 ($p = 0.124$), offering no evidence for an additional fixed-mass concentration shift.

Crowding, measure-difference, tidal-index, radial, and morphology-stratified size diagnostics are given in Appendix G.

4. Discussion: drivers of structural evolution in compact groups

4.1. An early-type-like satellite excess linked to dense group cores

The clearest residual difference between CG₄ and regular-group controls is morphological: a higher Galaxy Zoo 1 elliptical-vote fraction, carried primarily by satellites. The adjusted excess is largest against the luminosity-selected Control_{4B} population and true four-member RG₄ groups, and smallest against the projected cores Control_{4C}. Matched quartet composition follows the same ordering. Our within-host estimator refines the surrounding-host comparison of Montaguth et al. (2026b) by asking whether morphology differs beyond shared host and central location. The excess is clear without radius but not after radius adjustment, consistent with early-type-rich central occupation rather than an additional categorical CG₄ effect. Samples, morphology definitions, structural measurements, and estimates differ, so this refines rather than contradicts their result.

The scale of the morphology difference depends on what is held fixed. For example, restricting the individual-galaxy match to satellites increases the elliptical-fraction difference to 0.192, showing the dilution produced by BGGs. Present-day stellar mass may also encode part of a galaxy's environmental history, so fixed-mass estimates describe the remaining association rather than the total accumulated transformation. These qualifications narrow the physical claim without erasing it: the result is a satellite population shift associated with the dense configurations selected as compact groups.

The pairwise tidal-index experiment strengthens this interpretation while also narrowing it. For elliptical morphology, the pairwise complete-case baseline CG₄ odds ratio is 1.95 ($p < 10^{-4}$). After adding the projected pairwise tidal-index proxy, the CG₄ odds ratio decreases to 1.39 ($p = 0.031$), while the tidal-index term itself has odds ratio 1.42 ($p < 10^{-4}$). This behaviour is expected if compact groups are selecting galaxies in locally dense projected configurations where repeated close companions, rather than the catalogue label alone, are linked to the elliptical/early-type-like population.

This proxy uses projected separations, so it may be inflated by line-of-sight alignments and is entangled with compact-group selection itself. We therefore read the tidal-index result as a local projected-density or immediate-neighbour proxy, not as a time-dependent tidal-field measurement. The Control_{4C} comparison is stringent because Control_{4C} is also selected as a projected compact core around a BGG. The persistence of a CG₄–Control_{4C} morphology contrast after closest-neighbour exclusion, together with attenuation in the tidal-index model, is consistent with CG₄ selecting a more extreme part of the local-density distribution rather than establishing a separate causal channel.

This reading is consistent with compact groups tracing the high-density, high-interaction tail of group environments – some genuinely compact, others dense cores or substructures inside larger haloes – and with stronger structural signatures reported for non-isolated compact groups in recent S-PLUS work. The 6 isolated CG₄s are compatible with loose RG₄ groups in satellite early-type fraction (group-level permutation $p = 1.00$) and have a lower satellite early-type fraction than the rest of CG₄ ($p = 0.023$); this pattern is consistent with a larger-scale host contribution, but the very small isolated sample prevents a firm separation of explanations (Appendix E). The available M_{200c} is a host-halo abundance-matching mass rather than a harmonised quartet property; adding it to the adjusted models mildly attenuates the elliptical odds ratios and changes no conclusion. Stellar-population diagnostics may provide additional constraints on the origin of these morphological differences. In particular, stellar metallicity, age, and abundance ratios such as $[\alpha/\text{Fe}]$ could help distinguish differences in integrated enrichment history from differences in star-formation timescale; we defer such an analysis to future work.

4.2. Elliptical-vote morphologies, S0 degeneracy, and transition galaxies

We externally checked these operational classes against the deep-learning morphologies of Domínguez Sánchez et al. (2018). The cross-match contains 2436/3857 unique galaxies and thus validates only the DR7-era matched population. Coverage is 173/248 (69.8%) in CG₄ and 2263/3609 (62.7%) in deduplicated controls: it is not worse in CG₄, but neither rate establishes representativeness. Of the 942 matched GZ1 E galaxies, 92.1% have DS18 $T\text{Type} \leq 0$, 54.2% also have $P_{S0} > 0.5$, and 28.7% have $P_{S0} \geq 0.8$. The latter classifier-threshold S0-like fractions are 54.7% versus 54.2% for CG₄ and controls, and 24.4% versus 29.1% at the stronger cut. Thus CG₄ E galaxies are not preferentially S0-weighted; these conditional compositions do not causally decompose the environment excess.

Conversely, among 429 high-confidence S0s ($T\text{Type} \leq 0$, $P_{S0} \geq 0.8$), 62.9% fall in our E class and 24.5% in S. This split has a clear inclination dependence, consistent with face-on S0s looking smooth while inclined systems expose their discs. The E fraction is 56.8% among 37 CG₄ S0s and 63.5% among 392 controls (Fisher $p = 0.48$); the small CG₄ subset reveals no detectable environment-dependent mapping, rather than equivalence. Because DS18 P_{S0} separates S0 from E only within $T\text{Type} \leq 0$, and GZ1 was not designed for E/S0 decomposition, our E class is a smooth/early-type proxy with substantial S0 content, not specifically a classical-elliptical class.

This GZ1 caveat matters when comparing with recent S-PLUS studies of compact-group transition galaxies and R_e –Sérsic-index trends (Montaguth et al. 2023, 2025, 2026a,b). Their structural measurements and transition populations probe pathways; our control hierarchy tests which apparent CG₄ signatures survive across genuine four-member, luminosity-selected, and BGG-centred projected-core quartets.

4.3. Why the star-formation and size signals are weaker

The observed star-formation contrast is consistent with being carried largely by the differing vote-based morphology mix rather than by an additional conditional offset. At fixed vote-based morphology, the conditional quenched fractions do not differ detectably across samples ($P(Q|E) = 0.908, 0.903, 0.887$, and 0.850 for CG_4 , $Control_{4B}$, $Control_{4C}$, and RG_4 ; homogeneity $p = 0.39$). A Kitagawa decomposition leaves conditional terms consistent with zero: -0.017 $[-0.073, +0.038]$ versus $Control_{4B}$, $+0.002$ $[-0.054, +0.058]$ versus $Control_{4C}$, and $+0.050$ $[-0.035, +0.136]$ versus RG_4 (group-blocked 95% intervals); the underlying conditional quenched fractions are given in Table B.2. Thus the quenched odds-ratio pattern is consistent with the morphology-mix interpretation in this observational decomposition, with no additional quenching offset detected at fixed vote-based morphology.

The star-formation non-detections are also limited by the spectroscopy. CG_4 has more catalogued galaxies with a nearest neighbour below the 55-arcsec fibre-collision scale than the controls (19.4% versus 7.7%). $Control_{4C}$ is higher still (9.9%) because it is selected as a compact-core analogue. Additional close companions may therefore be preferentially missing in CG_4 , underrepresenting short-lived starbursts or post-starburst phases and biasing the pairwise tidal index downward. The MPA-JHU aperture corrections may also be less secure for interacting or blended members (Brinchmann et al. 2004; Kauffmann et al. 2003a). We therefore treat the sSFR, SFMS, and $H\alpha$ results as conservative secondary non-detections, not as proof of no star-formation effect.

The size test carries a parallel message. Coenda et al. (2012) found compact-group galaxies to be smaller at fixed luminosity than loose-group and field samples, especially for early types, using SDSS Petrosian half-light radii. We re-pose that question with the non-split HMCG-based CG_4 sample against three regular-group controls, seeing-corrected model sizes, a Petrosian cross-check, cluster-robust adjusted models, one-to-one matching, and crowding tests. Under this design, the pooled adjusted models do not show a compact-group size offset at fixed stellar mass relative to regular groups (satellite coefficient -0.010 dex, 95% interval including zero); the satellite fit bounds any mean CG_4 size deficit to roughly 0.04 dex. The seeing-corrected matched difference is likewise small (-0.019 dex). This is compatible with a compact-group versus field offset if ordinary-group cores have already undergone comparable structural processing. Two guardrails apply: a negative Δ at fixed mass can reflect tidal truncation or stripping, but also higher bulge dominance or residual blending in crowded fields. The morphology-stratified fits address the former, and the Petrosian cross-check, measure-difference diagnostic, and crowding exclusions bound the latter (Appendix G); future claims of compact-group size evolution should be read through these blocks together, not through a single pooled coefficient.

5. Conclusion

We have tested whether galaxies in CG_4 compact groups remain special after removing split systems, comparing them with RG_4 , $Control_{4B}$, and $Control_{4C}$ through the individual-galaxy, group-composition, and shared-host con-

trasts defined in Sect. 2.4. The inference applies to this luminous, volume-limited, exactly four-member non-split HMCG sample rather than to all compact groups.

The main result is an early-type-like satellite excess. CG_4 galaxies have a higher Galaxy Zoo 1 elliptical-vote fraction than all three regular-group comparisons after accounting for observed galaxy and group properties. The excess is strongest against $Control_{4B}$ and RG_4 and weakest against the projected cores $Control_{4C}$; matched quartet composition and satellite-mass-balanced variants show the same hierarchy. BGGs do not show a corresponding adjusted difference, and the preferred within-host analysis shows no clear excess beyond the central locations occupied by CG_4 members. Vote-threshold, continuous-vote, Sérsic-index, close-pair, and independent DS18 early/late checks support the morphology result. A projected tidal-density proxy attenuates the coefficient, consistent with local density contributing to the pattern, although the projected data do not identify mediation or a unique causal mechanism.

The star-formation result is weaker and consistent with being largely associated with the differing vote-based morphology mix, with no additional conditional offset detected at fixed vote-based morphology. CG_4 satellites have a lower raw star-forming fraction, especially relative to RG_4 , but adjusted evidence is control-dependent, matched contrasts provide no evidence for a quenched-fraction difference, and star-forming CG_4 galaxies show no robust main-sequence offset.

At fixed stellar mass, redshift, and group-scale covariates, the seeing-corrected satellite size estimate is -0.010 dex and its confidence interval includes zero; the matched Simard difference is -0.019 dex. The small Petrosian-only matched offset is not reproduced by the preferred size measure or across controls, so we do not claim a compact-group size effect.

The remaining secondary diagnostics do not identify a simple alternative driver: colours are control-sensitive and based on a biased matched subset, AGN-like fractions and control-specific magnitude gaps do not provide comparably robust parallel offsets, the $H\alpha$ diagnostic lacks age-sensitive spectral indices, and projected phase-space trends do not localize the contrast uniquely to the innermost satellites.

We therefore interpret CG_4 systems as hosting a population shifted towards smooth/early-type-like morphologies, not a uniquely classical-elliptical excess, consistent with accumulated structural transformation in dense group cores and/or preprocessing. This does not establish a compact-group-specific quenching mechanism at the present epoch, and the dynamically isolated subset is too small to isolate a distinct mode. Explicit E-S0-spiral and transition-galaxy classifications, bulge-disc structure, homogeneous halo masses, and age-sensitive stellar-population diagnostics will be needed to determine whether the satellite excess reflects disk fading, tidal heating, or merger-driven transformation.

Data availability

The input data are public SDSS, MPA-JHU, Galaxy Zoo, Simard et al. (2011), group, and compact-group catalogues cited above. Derived galaxy and group tables will be made available through CDS at publication. The analysis and paper-generation code is available at <https://github.com>.

[com/MatthieuTricottet/CG_Gals](https://www.sdss4.org/); the full SDSS query is in `src/data_loader.py`. The article release, including on-line diagnostic supplements, will be archived with a DOI at acceptance.

Acknowledgements. Funding for the Sloan Digital Sky Survey IV has been provided by the Alfred P. Sloan Foundation, the U.S. Department of Energy Office of Science, and the Participating Institutions. SDSS acknowledges support and resources from the Center for High-Performance Computing at the University of Utah. The SDSS web site is www.sdss4.org. SDSS is managed by the Astrophysical Research Consortium for the Participating Institutions of the SDSS Collaboration including the Brazilian Participation Group, the Carnegie Institution for Science, Carnegie Mellon University, Center for Astrophysics | Harvard & Smithsonian (CfA), the Chilean Participation Group, the French Participation Group, Instituto de Astrofísica de Canarias, The Johns Hopkins University, Kavli Institute for the Physics and Mathematics of the Universe (IPMU) / University of Tokyo, the Korean Participation Group, Lawrence Berkeley National Laboratory, Leibniz Institut für Astrophysik Potsdam (AIP), Max-Planck-Institut für Astronomie (MPIA Heidelberg), Max-Planck-Institut für Astrophysik (MPA Garching), Max-Planck-Institut für Extraterrestrische Physik (MPE), National Astronomical Observatories of China, New Mexico State University, New York University, University of Notre Dame, Observatório Nacional / MCTI, The Ohio State University, Pennsylvania State University, Shanghai Astronomical Observatory, United Kingdom Participation Group, Instituto Nacional de Astrofísica, México, University of Arizona, University of Colorado Boulder, University of Oxford, University of Portsmouth, University of Utah, University of Virginia, University of Washington, University of Wisconsin, Vanderbilt University, and Yale University. This publication uses Galaxy Zoo classifications; we thank the volunteers who made them possible. We used the MPA-JHU catalogues, the [Simard et al. \(2011\)](#) VizieR catalogue, and Astropy ([Astropy Collaboration et al. 2022](#)), Astropy ([Ginsburg et al. 2019](#)), NumPy ([Harris et al. 2020](#)), SciPy ([Virtanen et al. 2020](#)), pandas ([McKinney 2010](#)), Matplotlib ([Hunter 2007](#)), statsmodels ([Seabold & Perktold 2010](#)), and scikit-learn ([Pedregosa et al. 2011](#)).

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Appendix A: SDSS data query

For the SDSS calibration sample, we extract `sfr_tot_p50`, `specsfr_tot_p50`, and `lgm_tot_p50` from `galSpecExtra`, `h_alpha_eqw` from `galSpecLine`, and `p_el_debiased` and `p_cs_debiased` from `zooSpec`. In that query, these spectroscopic tables are joined to `SpecObj` by `specObjID`, while the photometric object is linked through `SpecObj.bestObjID=PhotoObj.objID`. The full result is split into the AGN-inclusive table and the non-AGN reference used for GMM calibration. The group samples instead inherit their DR12 `sfr_tot_p50`, `lgm_tot_p50`, and stored `specObjID` from the parent catalogues; only the Galaxy Zoo vote fractions are added to those rows by photometric `objid`. Their final $\log sSFR$ is the stored difference `sfr_tot_p50-lgm_tot_p50`.

We re-audited all 6076 final sample rows against `SpecObjAll` in DR12, retaining the stored `specObjID` even when the same photometric `objid` has multiple spectra. All 3855 unique non-zero identifiers resolve to the same photometric object; two `Control4B` rows have no stored spectrum identifier. The earlier low recovery resulted from looking up these DR12 identifiers in DR16: `specObjID` encodes the reduction version, so reprocessed BOSS observations can retain the same plate-MJD-fibre while acquiring a different identifier. On the exact DR12 observations, the missing rows comprise 4 `sfr_tot_p50` sentinels and 14 spectra without `galSpecExtra` coverage in `CG4`, 29 and 149 (plus the two rows without a spectrum identifier) in `Control4B`, 27 and 164 in `Control4C`, and 0 and 5 in `RG4`, respectively. `galSpecInfo` is complete for the sentinel subsets but absent for the much larger no-`galSpecExtra` subsets. The available cross-match is therefore insufficient to establish the origin of the missing $sSFR$ estimates.

The full query is provided in `src/data_loader.py` in the code repository (see Data availability).

Appendix B: sSFR classification details

We identify BPT-classifiable galaxies by requiring measured values of both emission-line ratios

$$\log_{10}([\text{N II}]\lambda 6584/\text{H}\alpha) \quad \text{and} \quad \log_{10}([\text{O III}]\lambda 5007/\text{H}\beta).$$

Among these galaxies, we flag AGN-like systems if they satisfy either

$$\log_{10} \frac{[\text{N II}]}{\text{H}\alpha} > 0,$$

or if they lie above the empirical demarcation of [Kauffmann et al. \(2003a\)](#):

$$\log_{10} \frac{[\text{O III}]}{\text{H}\beta} > \frac{0.61}{\log_{10}([\text{N II}]/\text{H}\alpha) - 0.05} + 1.3,$$

in which case they are excluded from the GMM calibration sample. This calibration flag is deliberately more inclusive than the three-class BPT scheme of Sect. 3.7, in which composite galaxies lie between the [Kauffmann et al. \(2003a\)](#) and [Kewley et al. \(2001\)](#) demarcations and only galaxies above the [Kewley et al. \(2001\)](#) line are labelled AGN-like, because its sole purpose is to keep possible AGN contamination out of the $sSFR$ calibration. Galaxies with missing diagnostic line ratios are BPT-unclassified, not proven non-AGN. They are retained for the broad $sSFR$ status assignment when valid SDSS stellar mass and $sSFR$ estimates are

Table B.1. Extreme-assignment bounds on the raw CG_4 -minus-control quenched-fraction difference. “Missing” columns give the missing- $sSFR$ percentage of each side; the observed difference uses classified galaxies only; min/max are the extreme corners with all missing galaxies on each side assigned to quenched or star-forming (all-galaxy denominators).

Scope	Control	Missing (%)		Δf_Q^{obs}	min	max
		CG ₄	ctrl			
All	Control _{4B}	7.3	6.4	0.025	−0.046	0.091
All	Control _{4C}	7.3	6.8	0.040	−0.034	0.107
All	RG ₄	7.3	2.2	0.174	0.115	0.210
Sat.	Control _{4B}	6.5	5.8	0.043	−0.021	0.101
Sat.	Control _{4C}	6.5	6.3	0.063	−0.005	0.122
Sat.	RG ₄	6.5	1.2	0.201	0.156	0.233

available, while the emission-line/AGN-like analysis treats BPT-unclassified objects separately. Reference-sample rows with unavailable `specsfr_tot_p50`, and group-sample rows with unavailable `sfr_tot_p50` or `lgm_tot_p50`, including the -9999 sentinel, have no valid $sSFR$ estimate; they are treated as missing data throughout: excluded from the GMM calibration where applicable, never assigned an $sSFR$ class, omitted from every $sSFR$ figure, and reported as counts only (Sect. 2.2). Galaxies with a measured $sSFR$ are divided into quenched and star-forming classes with a two-component Gaussian Mixture Model (GMM) in the $\log M_\star$ - $\log sSFR$ plane ([McLachlan & Peel 2000](#)). The calibration excludes galaxies flagged as AGN-like where the required BPT line ratios are available, while BPT-unclassified galaxies with valid SDSS stellar mass and $sSFR$ estimates can still receive a broad $sSFR$ class. The quenched/star-forming boundary is the locus where the two component posterior probabilities are equal; the resulting classification for the SDSS non-AGN reference is shown in Fig. B.1. A fixed-threshold robustness check confirms that the qualitative hierarchy is not tied to the exact GMM split (Sect. 3.7).

Extreme-assignment bounds for the missing class. Because the missing- $sSFR$ class could in principle hide differential quenching, Table B.1 bounds every raw quenched-fraction contrast by the four extreme assignments in which *all* missing- $sSFR$ galaxies on *each* side are set to quenched or to star-forming, with consistent all-galaxy denominators on both sides (so CG_4 is never compared with a classified-only control fraction). The differential missingness is negligible against `Control4B` and `Control4C` (7% of CG_4 members versus 6% and 7%) and material only against `RG4` (2%). The raw RG_4 contrast survives all four corners (all-member difference between 0.12 and 0.21 around the observed 0.17; satellites 0.16 to 0.23), whereas the small `Control4B` and `Control4C` differences can change sign only under the doubly adversarial corner and are not the basis of any claim in this paper.

Star formation at fixed morphology. Table B.2 gives the quenched fractions among $sSFR$ -classified galaxies at fixed vote-based morphology. The four samples are statistically compatible at fixed morphology (homogeneity of $P(Q|E)$: $\chi^2 p = 0.39$); the Kitagawa decomposition quoted in Sect. 4.3 splits each raw CG_4 -minus-control quenched-fraction offset into its morphology-mix and conditional parts on this basis, with group-blocked bootstrap intervals (4000 draws).

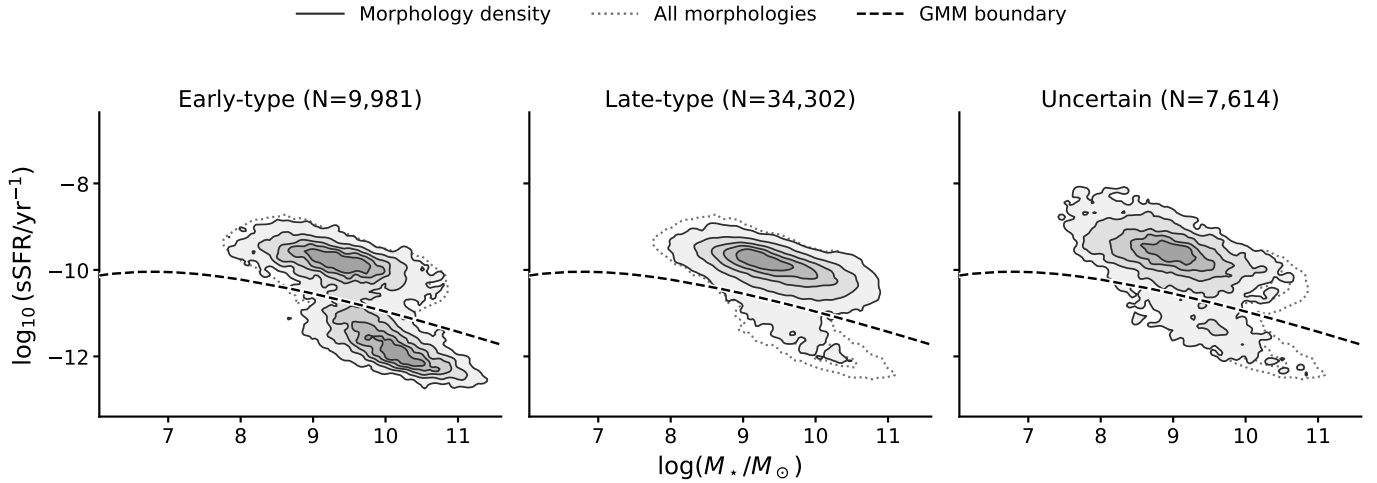


Fig. B.1. Galaxy Zoo morphology in sSFR–stellar-mass space for the SDSS non-AGN reference, excluding galaxies without sSFR estimates. Panels show early-type, late-type, and uncertain morphologies; no unavailable morphologies occur in this reference sample ($N = 0$). Solid class-specific contours enclose 20, 40, 60, 80, and 95% of the density, grey dotted contours show the full reference distribution at the same enclosed-probability levels, and the dashed curve is the unchanged adopted star-forming–quenched GMM boundary.

Table B.2. Conditional quenched fractions $P(Q|\text{morphology})$ among sSFR-classified galaxies, with Wilson 95% intervals.

	CG ₄	Control _{4B}	Control _{4C}	RG ₄
Elliptical	0.908 ^{+0.040} _{-0.065}	0.903 ^{+0.016} _{-0.019}	0.887 ^{+0.017} _{-0.020}	0.850 ^{+0.069} _{-0.111}
Spiral	0.326 ^{+0.105} _{-0.090}	0.357 ^{+0.027} _{-0.026}	0.328 ^{+0.027} _{-0.026}	0.295 ^{+0.084} _{-0.072}

Notes. For the uncertain-morphology class, $P(Q|U) = 0.429, 0.552, 0.594$, and 0.267 ($n = 7, 116, 143$, and 15 classified galaxies, same sample order).

Table C.1. CG₄ galaxies inside would-be Control_{4C} quartets of the parent catalogue, by compact-group class.

CG ₄ class	CG ₄ groups	Lim groups	Galaxies
Embedded	37 / 37	37	145
Isolated	6 / 6	6	24
Predominant	14 / 19	14	52
Split	4 / 16	4	12

Notes. “CG₄ groups” counts compact groups with at least one member inside a parent-group quartet, over the total number of compact groups of that class (split groups included, as in the exclusion).

Appendix C: Compact groups as projected cores of ordinary groups

The compact-group exclusion of Sect. 2.1 is not a technicality: for 61 of the 765 parent Lim groups, the would-be Control_{4C} quartet (BGG plus the three closest projected companions within 3 magnitudes) contains at least one CG₄ galaxy — 233 CG₄ galaxies in total. Table C.1 breaks this overlap down by the Zheng & Shen (2021) embeddedness class: essentially every Embedded and Predominant compact group *is* the projected core of an ordinary group. The Isolated entries are trivial self-overlap: for each of the 6 isolated systems, the overlapping Lim group is the compact group’s own host, whose BGG and three nearest projected companions are exactly the four CG₄ members. Had these groups not been excluded, the same galaxies would have appeared on both sides of the comparisons, biasing every contrast toward null.

Appendix D: Star-forming main-sequence details

We model the star-forming main sequence as a function of stellar mass using polynomial orders 1–4; the polynomial-order comparison is provided in the online supplement. The diagnostic shown there is the residual RMS scatter, not a formal reduced χ^2 , because no per-galaxy uncertainty model is used. We adopt order 2 because it is the lowest-order model with the smallest five-fold cross-validated RMS

scatter (0.305 dex, compared with 0.306 dex for order 1). We then use order 2 to compute the residual of each star-forming galaxy with respect to the main sequence. A positive residual means higher sSFR at fixed stellar mass relative to the fitted star-forming main sequence.

In the comparisons below, Δ is the CG₄-minus-control median residual; positive values place CG₄ star-forming galaxies above the fitted main sequence relative to the control. Each quoted interval is the central 68% group-bootstrap interval on the median difference, and each p is the corresponding two-sided group-bootstrap sign-crossing probability (10 000 resamples).

The median main-sequence residual of CG₄ galaxies is:

- Relative to Control_{4B}, the median offset is $\Delta = 0.056$ ($0.027 \leq \Delta \leq 0.098$, 16th–84th percentile bootstrap interval), with $p = 0.064$.
- Relative to Control_{4C}, the median offset is $\Delta = 0.058$ ($0.039 \leq \Delta \leq 0.103$, 16th–84th percentile bootstrap interval), with $p = 0.030$.
- Relative to RG₄, the median offset is $\Delta = 0.046$ ($-0.043 \leq \Delta \leq 0.098$, 16th–84th percentile bootstrap interval), with $p = 0.589$.

Because the raw residual comparison and the matched comparison condition on different covariate structures, and

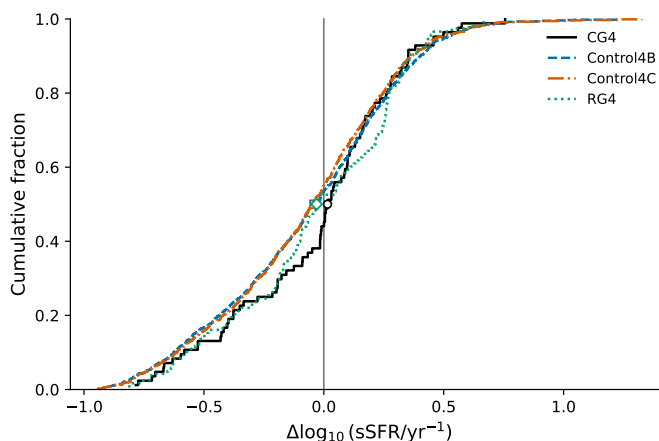


Fig. D.1. Cumulative distributions of the existing order-2 SFMS residuals for star-forming galaxies. Zero marks the adopted SFMS; symbols mark sample medians. The normalisation gives each sample equal visual weight.

because the apparent offset changes in strength after matching, we treat SFMS residuals as a secondary diagnostic rather than as evidence for a robust compact-group star-formation offset (Sect. 3.1).

The full residual distributions overlap closely (Fig. D.1). In particular, the CG₄ distribution shows neither a pronounced negative-residual tail of anaemic star formers nor a distinct enhanced tail. This visual diagnostic leaves the median-offset interpretation unchanged.

Within CG₄, the star-forming main-sequence offsets of the Embedded, Isolated and Predominant classes remain statistically consistent with one another.

Appendix E: Environmental, group-scale, and colour diagnostics

This appendix collects the group-scale and colour diagnostics summarized in Sect. 3.7.

E.1. Distance to the BGG

We compare the sSFR of satellite galaxies with their distance to the BGG, expressed both in projected kpc and in units of $\langle R_{ij} \rangle$.

The following sample-specific monotonic trends have small p -values.

In CG₄, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_s = 0.24$ for 174 satellites ($p = 1.5 \times 10^{-3}$) and Pearson coefficient $r = 0.21$ ($p = 5.5 \times 10^{-3}$).

In Control_{4B}, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_s = 0.08$ for 1972 satellites ($p = 3.2 \times 10^{-4}$) and Pearson coefficient $r = 0.08$ ($p = 1.9 \times 10^{-4}$).

In Control_{4C}, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_s = 0.19$ for 1977 satellites ($p < 10^{-4}$) and Pearson coefficient $r = 0.16$ ($p < 10^{-4}$).

In Control_{4C}, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_s = 0.11$ for 1977 satellites ($p < 10^{-4}$) and Pearson coefficient $r = 0.11$ ($p < 10^{-4}$).

E.2. Dominance

We split groups into dominated systems, where the BGG contributes more than 60% of the group luminosity, and non-dominated systems. Because this appendix screens many group properties, p_{adj} denotes the exploratory-battery adjustment defined in Sect. 2.4; Cliff's δ is the rank-based effect size. The split does not produce a strong compact-group signal by itself.

Across the control and reference samples, the clearest binary separation is instead in group luminosity. Dominated systems have lower L_{group} than their non-dominated counterparts: Control_{4B} gives 9.1×10^{10} versus $1.28 \times 10^{11} L_{\odot}$ ($p_{\text{adj}} < 10^{-4}$, $\delta = -0.55$); Control_{4C} gives 9.66×10^{10} versus $1.07 \times 10^{11} L_{\odot}$ ($p_{\text{adj}} < 10^{-4}$, $\delta = -0.25$); and RG₄ gives 7.09×10^{10} versus $9.82 \times 10^{10} L_{\odot}$ ($p_{\text{adj}} = 1 \times 10^{-4}$, $\delta = -0.59$). The negative δ values mean that dominated groups tend to occupy the lower-luminosity side of each comparison.

The spiral-fraction correlations show a broader environmental pattern rather than a clean dominated/non-dominated dichotomy. The most coherent set appears in Control_{4C}, where S_{frac} increases with $\langle R_{ij} \rangle$ in both dominated and non-dominated groups ($\rho_s = 0.30$, $p < 10^{-4}$ and $\rho_s = 0.26$, $p < 10^{-4}$), while weaker negative trends are found with σ_v and L_{group} . Control_{4B} contributes only weaker non-dominated-group correlations. In CG₄, the reported trends occur in small subsets, with S_{frac} decreasing with L_{group} among dominated groups ($\rho_s = -0.40$, 36 groups) and with $\Delta V_{\text{BGG}}/\sigma_v$ among non-dominated groups ($\rho_s = -0.45$, 24 groups); we therefore treat these as supporting diagnostics rather than the main dominance result. The full distributional diagnostics are provided in the online supplement.

E.3. Morphology and domination

When we further split the samples by domination and by the morphology of the BGG, the following cases have small p -values.

- Control_{4B} dominated groups, where the median satellite spiral fraction is 1.00 for spiral BGGs and 0.67 for elliptical BGGs. The association between BGG morphology and satellite spiral fraction has a group-level odds ratio of 1.842 ($p = 0.01$) in this sample; the same within-sample contrast under a satellite-level clustered uncertainty treatment gives an odds ratio of 1.842 ($p = 0.02$).
- Control_{4C} dominated groups, where the median satellite spiral fraction is 0.67 for spiral BGGs and 0.67 for elliptical BGGs. The association between BGG morphology and satellite spiral fraction has a group-level odds ratio of 1.972 ($p < 10^{-4}$) in this sample; the same within-sample contrast under a satellite-level clustered uncertainty treatment gives an odds ratio of 1.972 ($p < 10^{-4}$).

In each case the group-level and satellite-level clustered treatments share the same point estimate by construction and differ only in the uncertainty model, which is why the odds ratios coincide.

E.4. Morphology versus Zheng–Shen class, magnitude gap, and BGG dominance

We next test whether the E/Sp morphology mix is associated with three group-level descriptors: the Zheng–

Table E.1. Elliptical and spiral counts by Zheng–Shen class for the CG₄ sample. Finite-vote uncertain morphologies and rows without Galaxy Zoo vote fractions are excluded. Split systems have zero counts because they were removed by the existing non-split CG₄ sample definition.

Zheng–Shen class	All CG ₄ E/Sp	CG ₄ BGG E/Sp	CG ₄ sat. E/Sp
Split	0/0	0/0	0/0
Isolated	7/11	3/1	4/10
Predominant	41/22	11/5	30/17
Embedded	76/53	22/11	54/42

Table E.2. Elliptical fractions $f_E = N_E/(N_E + N_{Sp})$ and median host halo mass by Zheng–Shen class, with Wilson 95% intervals.

Class	Groups	f_E (all)	f_E (sat.)	$\log M_{200c}$
Isolated	6	$0.389^{+0.225}_{-0.186}$	$0.286^{+0.261}_{-0.169}$	12.86
Embedded	37	$0.589^{+0.081}_{-0.086}$	$0.562^{+0.095}_{-0.100}$	13.13
Predominant	19	$0.651^{+0.106}_{-0.123}$	$0.638^{+0.122}_{-0.143}$	13.83

Notes. Host $\log M_{200c}$ is the Yang–Lim abundance-matching host-halo mass (Appendix F.3); for isolated systems it describes the quartet itself.

Shen compact-group class, the first–second magnitude gap $\Delta m_{12} = M_{r,2} - M_{r,1}$, and the continuous BGG luminosity fraction $f_{L,BGG} = L_{BGG}/L_{group}$. These tests use only the existing Galaxy Zoo elliptical and spiral classes; finite-vote uncertain morphologies and rows without Galaxy Zoo vote fractions are excluded from the binary fits. We run the tests separately for all galaxies, BGGs only, and satellites only, so that any trend driven by the BGG itself is not folded into the satellite result. For satellites, $f_{L,BGG}$ mechanically includes the satellite luminosity in the group-luminosity denominator, so any satellite-level trend with $f_{L,BGG}$ is partly definitional; we therefore report the satellite models with a stellar-mass control.

The magnitude-gap and BGG-luminosity-fraction quantities are available for all 118 CG₄+RG₄ groups used in this test. The Zheng–Shen class is available only for compact groups. Split systems are absent from the current CG₄ sample because they are removed before the paper analysis, and no corresponding Zheng–Shen class is defined for the RG₄ catalogue used here. We therefore report the RG₄ class test as unavailable, while retaining RG₄ in the Δm_{12} and $f_{L,BGG}$ comparisons.

Overall, these tests provide little evidence that the E/Sp mix is driven by Zheng–Shen class, magnitude gap, or BGG luminosity fraction. The few weak associations are treated as descriptive diagnostics (Tables E.1 and E.3).

For the Zheng–Shen compact-group class, the CG₄ associations are weak, including the satellite-only test. Because the class is not defined for the RG₄ catalogue, no RG₄ class comparison is attempted.

The class structure of the early-type fraction itself is summarized in Table E.2. Group-level permutation tests on the difference of per-group mean satellite f_E (20 000 permutations, two-sided, add-one convention) give $p = 0.023$ for Isolated versus the rest of CG₄, $p = 1.00$ for Isolated versus RG₄, and $p = 0.40$ for Isolated versus Control_{4C}: the isolated systems are compatible with the loose RG₄ groups and sit below the embedded classes of their own sample.

The Δm_{12} , $f_{L,BGG}$, and CG₄–RG₄ interaction coefficients are all weak across the all-galaxy, BGG, and satellite fits.

Table E.3. Tests of the E-vs-Sp morphology mix against compact-group class, magnitude gap, and BGG luminosity fraction. The continuous columns report the available adjusted logistic model for $P(E)$; all-galaxy and satellite fits use group-clustered standard errors. Odds ratios are reported per magnitude for Δm_{12} and per 0.1 increase in luminosity fraction for $f_{L,BGG}$.

Subset	Sample	Test	Result
all galaxies	CG ₄	Class	$p = 0.137$
		Δm_{12}	OR 1.42, $p = 0.167$
		$f_{L,BGG}$	OR 1.29, $p = 0.080$
all galaxies	RG ₄	Class	not defined
		Δm_{12}	OR 1.33, $p = 0.220$
		$f_{L,BGG}$	OR 1.09, $p = 0.540$
BGGs only	CG ₄	Class	$p = 1.000$
		Δm_{12}	OR 1.77, $p = 0.426$
		$f_{L,BGG}$	OR 1.68, $p = 0.174$
BGGs only	RG ₄	Class	not defined
		Δm_{12}	OR 1.58, $p = 0.312$
		$f_{L,BGG}$	OR 1.21, $p = 0.520$
satellites only	CG ₄	Class	$p = 0.066$
		Δm_{12}	OR 1.28, $p = 0.346$
		$f_{L,BGG}$	OR 1.18, $p = 0.321$
satellites only	RG ₄	Class	not defined
		Δm_{12}	OR 1.41, $p = 0.342$
		$f_{L,BGG}$	OR 1.12, $p = 0.584$

A class interaction is undefined because the Zheng–Shen compact-group class has no RG₄ counterpart, and no result survives the exploratory FDR screen.

E.5. Crossing time

Group radii and t_{cross} use projected proper separations from the Planck 2015 angular-diameter distance, correcting the luminosity-distance convention used for this diagnostic in Tricottet et al. (2025); the values decrease by 3.3–9.6% (median 7.2%), without changing the reported correlations at quoted precision. The secondary colour audit places the CG₄ satellite blue-fraction correlation at $\rho_S = 0.365$ (adjusted $p = 0.05003$); this weak boundary result does not change the exploratory interpretation of the colour diagnostics.

The crossing-time diagnostic mainly separates systems by dynamical scale rather than by luminosity alone. Shorter t_{cross} systems have larger virial mass-to-light ratios and virial masses: the correlations with $\log(M_{VT}/L_r)$ and $\log(M_{VT})$ are $\rho_S = -0.39$ ($p < 10^{-4}$) and $\rho_S = -0.33$ ($p < 10^{-4}$), respectively, over 816 groups. By contrast, the relation with total group luminosity is formally detected but very weak ($\rho_S = 0.08$). We therefore read t_{cross} as a dynamical compactness indicator tied primarily to virial mass and mass-to-light ratio, with little independent leverage from L_{group} .

E.6. Optical colours

The retained colour-analysis matches include 110 of 248 galaxies in CG₄, 931 of 2792 galaxies in Control_{4B}, 1153 of 2812 galaxies in Control_{4C}, 102 of 224 galaxies in RG₄. This stricter subset differs from the availability audit in Fig. F.2: the figure’s colour row only requires the four derived SDSS colour columns to be present in the final catalogue, giving 217 of 248 CG₄ galaxies, whereas the colour analysis also requires successful matched-photometry retention and the covariates used below. The fractions retained in this

stricter colour-analysis subset are 44.4% in CG4, 33.3% in Control4B, 41.0% in Control4C, 45.5% in RG4.

The raw catalogue colour-mass fits indicate that CG₄ does not simply follow the same colour-mass relation as every comparison sample: common slopes are rejected in all four colours, with the directional slope differences having small adjusted p-values mainly in the bluer colours and depending on the adopted comparison catalogue. In the satellite-only comparison, the fitted offsets and slopes likewise depend on the control definition. These tests motivate the robustness analysis rather than establishing a stand-alone compact-group colour effect.

We refit satellite colours with cluster-robust standard errors. In the pooled mass- and redshift-adjusted comparison, the negative CG₄ coefficients are weak. Adding morphology and sSFR class does not reveal a clear negative coefficient. These coefficients remain control-sample dependent, with non-uniform split-sample behaviour, so they do not identify a global colour shift or a simple interaction-triggered star-formation population.

Residual colour trends with projected BGG-centric distance are consistent with stronger processing near the group centre and a less processed outer population, but these trends use the same biased colour subset and are treated as exploratory. The group-level BGG-satellite colour concordance is similarly secondary: redder BGGs tend to have redder mean satellite populations in the combined group sample, but the CG₄ slope is not a robustly distinct population-level signature. Detailed colour-mass, satellite, and radial diagnostics are provided in the online supplement. Figure E.1 summarizes the model and control-sample dependence that motivates our interpretation: the present data do not establish a robust global blue or red offset for CG₄ satellites.

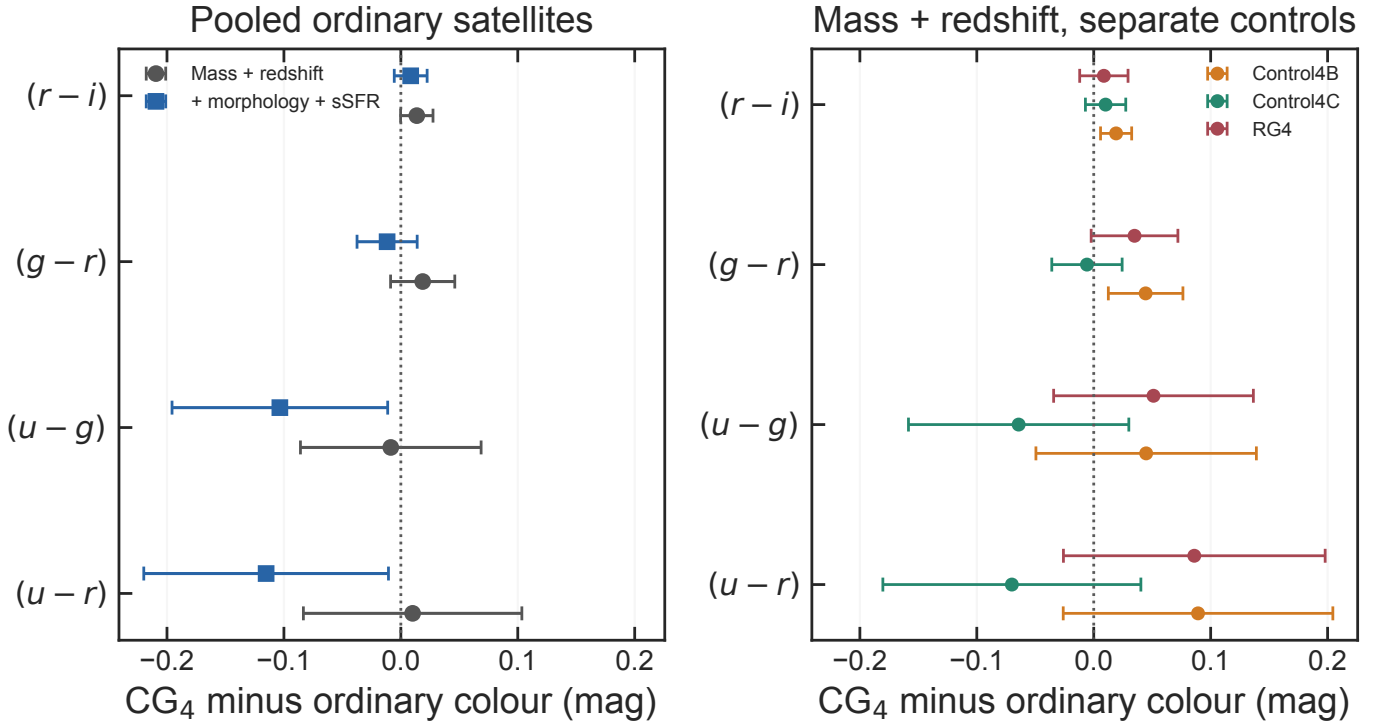


Fig. E.1. Robustness of the satellite colour coefficients. Left: pooled ordinary-group comparison before and after controlling for morphology and sSFR class. Right: mass- and redshift-adjusted comparisons against each control catalogue separately. Points show the CG_4 minus ordinary coefficient and bars show 95% confidence intervals; negative values indicate bluer CG_4 satellites.

Appendix F: Pooled models, matching details, and selection diagnostics

F.1. Pooled adjusted models

As a compact summary across the heterogeneous control definitions we also fit binomial models to the pooled catalogue. Because the control samples overlap heavily, the pooled control side is first deduplicated by SDSS objid and standard errors are clustered by physical Lim group. The common adjustment set includes stellar mass, redshift, satellite/BGG rank, group luminosity, and velocity dispersion. This pooled result does not replace the physically interpretable per-control contrasts.

For quenched status, the CG_4 odds ratio is 1.35, with 95% confidence interval [0.98, 1.86] and $p = 0.068$.

For quenched satellite status, the CG_4 odds ratio is 1.53, with 95% confidence interval [1.10, 2.11] and $p = 0.011$.

For elliptical morphology, the CG_4 odds ratio is 1.77, with 95% confidence interval [1.32, 2.37] and $p = 1.4 \times 10^{-4}$.

For elliptical satellite morphology, the CG_4 odds ratio is 2.08, with 95% confidence interval [1.51, 2.88] and $p < 10^{-4}$.

For elliptical BGG morphology, the CG_4 odds ratio is 0.78, with 95% confidence interval [0.39, 1.53] and $p = 0.464$.

The pooled coefficients reinforce the satellite morphology shift. The quenched estimate is treated more cautiously because it changes across the separate controls and is not reproduced by the matched contrast.

F.2. Galaxy-level matching details

The galaxy-level control pool of Sect. 3.3 is deduplicated by SDSS objid before matching: 5828 control rows collapse to 3609 unique galaxies. The 14 unmatched CG_4 galaxies were excluded because they lay outside the common propensity-score support or exceeded the adopted caliper. By kept label, the matched controls comprise 159 Control4B, 57 Control4C, 18 RG4; by physical provenance, 18 of the 234 matched controls are physically RG4 galaxies (which also appear in Control4B through the sample overlap), and the full per-galaxy provenance is recorded in the released `matched_control_provenance.csv`. The covariate balance before and after matching is shown in Fig. F.1; the per-outcome matched-effect figure is provided in the online supplement.

The 234 matched controls occupy 204 physical Lim groups, and 26 control hosts link more than one treated CG_4 group. We therefore resample the 33 connected components of the treated-group/control-host graph. The elliptical-fraction difference remains 0.139 (95% interval [0.051, 0.244]; $p = 0.004$), and the quenched-fraction difference is 0.031 ([−0.090, 0.120]; $p = 0.649$). The analogous matched-size sensitivities are −0.019 dex for Simard radius ($p = 0.310$), −0.033 dex for Petrosian radius ($p = 0.016$), and −0.008 for concentration ($p = 0.896$). The one-proxy Petrosian result warrants the same caution as the original size analysis.

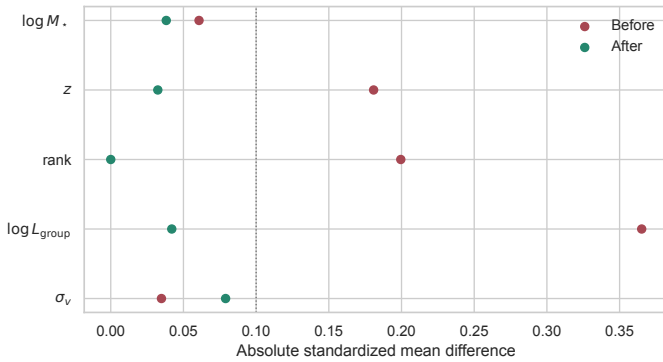


Fig. F.1. Absolute standardized mean differences before and after matching.

F.3. Selection diagnostics

The fractions with complete measurements for the major derived quantities are shown in Fig. F.2; this consolidated audit includes the Simard and Petrosian size availability used by Sect. 3.8. The colour-matched versus unmatched comparisons quantify shifts in stellar mass, redshift, rank, star-formation class and morphology.

The median nearest-neighbour angular separation is 100.9 arcsec in CG₄ and 255.9 arcsec in the controls; the fractions below the 55-arcsec SDSS fibre-collision scale are 19.4% and 7.7%, respectively.

Under the availability definitions shown in Fig. F.2, no major availability difference has both an adjusted p-value below 0.05 and a difference exceeding 10 percentage points.

The group-scale availability row is based on the group covariates available for this catalogue: radius scale, velocity dispersion, group luminosity, and BGG luminosity dominance. The tables do carry an M_{200c} estimate, but it is the Yang–Lim abundance-matching M_{180m} of the identified *host* halo (converted as in Paper I, Appendix A), not a quartet property: for embedded quartets it describes the richer host while for isolated systems it describes the quartet itself, it derives from group luminosity by abundance matching and is therefore nearly collinear with the group-luminosity covariate already in the models, and it is entangled with the compact/loose selection. We therefore exclude it from the harmonised covariate set. A sensitivity fit adding $\log M_{200c}$ to the published covariates leaves every elliptical conclusion unchanged: the pooled odds ratio moves from 1.77 to 1.62 ($p = 0.0020$), and the per-control odds ratios become 2.00, 1.56, and 2.65 against Control_{4B}, Control_{4C}, and RG₄ (all $p < 0.005$).

Restricting the pairwise tidal index of Sect. 3.6 to quartet co-members is likewise innocuous: recomputing T_i with all spectroscopic Lim members of the identified host group included (positions from the group catalogue; stellar masses from MPA–JHU where available, so the host-inclusive index is itself a lower bound) shifts $\log T_i$ by a median of 0.009 dex (sample medians ≤ 0.02 dex, and identically zero for RG₄, whose groups have exactly four members), with rank correlation 0.91 between the two definitions — the residual reordering is confined to a minority of loose quartets embedded in rich hosts — and refitting the with-index elliptical model on the host-inclusive index moves the attenuated CG₄ coefficient only from 1.39 to 1.33. T_i should thus be read as the local, galaxy-scale tidal density traced by cat-

alogued galaxies; any smooth host-halo contribution is not measured by this sum and is probed separately through the host-mass sensitivity fit above and the within-host analysis of Sect. 3.4.

The colour-selection mass audit is provided in the online supplement.

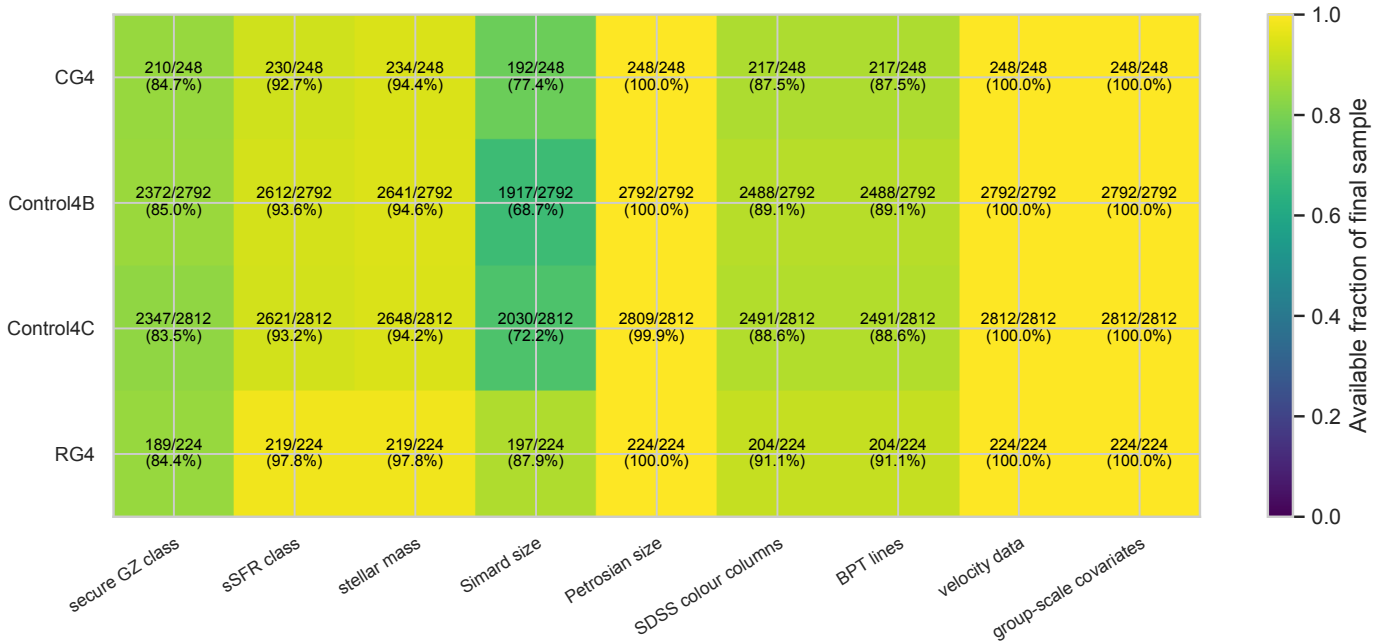


Fig. F.2. Availability fractions for the major derived quantities in each sample, including the Simard and Petrosian size-availability columns used by the size analysis. Cell labels give the number available over the final per-sample galaxy count and the corresponding percentage. The colour row records completeness of the four derived SDSS colour columns in the final catalogue, rather than the stricter colour-analysis matched subset used in Appendix E.6. The Simard-size column reflects the DR7 bridge, redshift, quality-window, blending, and pegged-Sérsic-index cuts described in Sect. 2.3.

Appendix G: Extended size diagnostics

The size-availability counts are included in the consolidated audit of Fig. F.2. This appendix therefore focuses on the extended size-effect diagnostics; the descriptive mass-size relations and the per-sample sizes predicted at $\log(M_\star/M_\odot) = 10.3$ are provided in the online supplement.

The crowding checks mirror Sect. 3.5: excluding galaxies with a projected neighbour within 55 arcsec leaves a satellite coefficient of -0.013 dex ($p = 0.463$), and adding the close-neighbour flag as a covariate gives -0.011 dex ($p = 0.479$).

The per-galaxy difference between the two measures, $d = \log R_{\text{chl},r} - \log R_{50,r}$, averages 0.054 dex, and is larger for elliptical galaxies (0.074 dex) than for spiral galaxies (0.042 dex), as expected if the Petrosian aperture preferentially truncates concentrated profiles. Its rank correlation with the nearest-neighbour separation is $\rho_s = 0.012$ ($p = 0.524$), so residual blending does not drive a strong measure-dependent size systematic.

Adding the pairwise projected tidal index of Sect. 3.6 to the all-galaxy size model changes the CG_4 coefficient from -0.012 dex to 0.005 dex on the pairwise complete-case sample; the local projected-density proxy attenuates the size coefficient in the same way it attenuates the morphology coefficient.

The radial diagnostic is exploratory: mass- and redshift-adjusted satellite size residuals correlate with projected BGG-centric distance with $\rho_s = 0.17$ ($p = 0.026$) in CG_4 and $\rho_s = 0.18$ ($p < 10^{-4}$) in the pooled controls. A more negative inner-region residual would be the tidal-truncation expectation; the observed trends are similar in CG_4 and the controls, so the data do not isolate a CG_4 -specific inner-region size suppression.

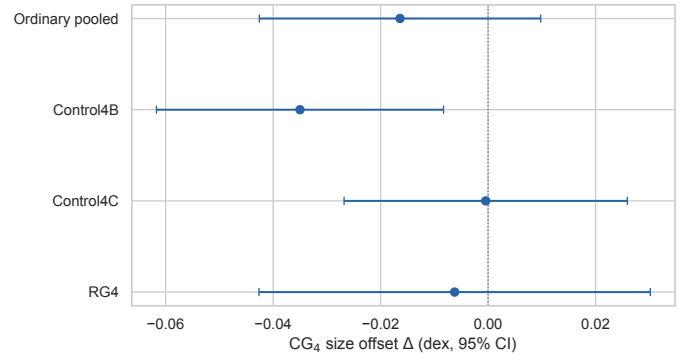


Fig. G.1. Mass- and redshift-adjusted CG_4 size coefficients against the pooled controls and against each control catalogue separately. Bars show 95% confidence intervals; negative values mean smaller CG_4 sizes at fixed mass.

Within morphology classes, the satellite-only fits give 0.036 dex for elliptical satellites ($p = 0.068$) and -0.005 dex for spiral satellites ($p = 0.831$), and the $\text{CG}_4 \times \text{morphology}$ interaction is 0.032 dex ($p = 0.348$). These strata are secondary diagnostics of whether size carries information beyond the morphology shift.